

Team Number :	apmcm25301061
Problem Chosen :	C

2025 APMCM summary sheet

This study provides a comprehensive quantitative assessment of the multi-dimensional impacts of the U.S. "reciprocal tariff" policy implemented in April 2025 on global trade flows and regional economic landscapes. Utilizing a suite of mathematical and econometric models, the research analyzes sector-specific and macroeconomic effects across different countries and industries.

In response to Question 1, we employed a Dirichlet Regression Model to analyze the trade share of soybean imports to China from the U.S., Brazil, and Argentina. The model effectively captured the significant shift in market share following the tariff policy and China's retaliatory measures, predicting a long-term equilibrium where U.S. market share diminishes to zero while Brazil and Argentina stabilize at approximately 55% and 45% respectively.

For Question 2, a Multivariate Logit Model was constructed to analyze the market share of Japanese automobiles in the U.S. across three channels: direct exports, U.S. production, and third-country production. This model quantifies how tariff adjustments will drive a strategic shift towards localized production, increasing it to approximately 115% of its original level, thereby explaining how the US-Japan automobile trade structure will change.

In addressing Question 3, an Interrupted Time Series (ITS) model combined with a SARIMAX model was used to evaluate the impact of U.S. tariff policies on the semiconductor manufacturing industry and trade. The result figure revealed differentiated effects on high-end, mid-range, and low-end chips, highlighting trade-offs between economic efficiency and national security.

Regarding Question 4, a Holt-Winters exponential smoothing model, enhanced with a piecewise elasticity adjustment, was developed to forecast U.S. import volumes and tariff revenues. The predictions indicate that tariff revenue will surge in the short term, followed by a medium-term decline of approximately 20% of the short-term peak due to declining import volumes. This allows us to analyze the net fiscal impact in the future two years.

Finally, for Question 5, a VARX model and dynamic equations were employed to simulate the short- and medium-term impacts of tariff adjustments on the U.S. economy. The study selected key financial and industrial indicators to assess the policy's effectiveness in driving manufacturing reshoring, concluding that the protective effect is short-lived and insufficient for sustainable industrial recovery.

The strengths and limitations of the models are evaluated, and potential improvements are suggested. The findings indicate that while the tariff policy yields short-term fiscal and protective benefits, its long-term structural costs are substantial, undermining global supply chains and failing to foster enduring manufacturing competitiveness.

Keywords: Tariff Policy Impact, multivariate logit model, time series analysis, Holt-Winters exponential smoothing, VARX model

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I. Introduction

1.1 Background

In April 2025, the United States implemented an aggressive tariff policy under the guise of "reciprocity," marking a fundamental shift in its trade strategy. The core of this policy included two measures: establishing a 10% "minimum benchmark tariff" on all trading partners and imposing additional tariffs on approximately 60 countries with trade deficits, including China. Its publicly stated goal was to correct so-called "unfair trade practices" and reverse the decline in U.S. manufacturing and the persistent trade deficit.

This move triggered a dramatic change in the global trade environment. On the one hand, tariff barriers distorted trade flows and altered bilateral trade balances in the short term; on the other hand, it inevitably provoked retaliatory measures from major trading partners, creating a cycle of escalating trade frictions, leading to a significant decline in both U.S.-China trade and imports. The adverse effects of high tariffs on U.S. trade and the economy cannot be ignored.

Therefore, in-depth quantitative analysis of the multidimensional impacts of this round of tariff adjustments becomes crucial. This not only involves short-term profit and loss assessments of the U.S. economy itself, such as trade deficits, industrial performance, and consumer prices, but also concerns the long-term structural impacts on other regional economic sectors, and the international trade order.

This study aims to construct a comprehensive analytical framework based on U.S.-regional trade data from recent years to assess the multi-faceted impact of U.S. tariff policies on trade flows and regional economic landscapes. Through analysis of the impact mechanisms, we seek to develop a scientific model capable of dynamically predicting future policy evolution trends, thereby providing empirical evidence and forward-looking recommendations for governments and businesses worldwide to address trade uncertainties and optimize strategic decision-making.

1.2 Given conditions

(1) The United States will implement a reciprocal tariff policy starting April 2, 2025, including: imposing a 10% minimum benchmark tariff on all trading partners and imposing additional tariffs on approximately 60 countries and regions with which it has a trade deficit.

(2) Changes in tariff levels: The weighted average tariff rate for all U.S. products will increase from 2.44% at the beginning of 2025 to 20.11% in August 2025.

(3) Relevant data and materials to support are provided in each sub-question.

1.3 Question restatement

Question 1. Analyze the current trade situation between China's soybean imports and the soybean exports of the United States, Brazil, and Argentina. Develop a model to assess the impact of U.S. tariff adjustments on the soybean industries of the U.S., Brazil, and Argentina. Based on the model, estimate the distribution of soybean export volume and value of the three countries after the adjustment.

Question 2. Analyze the current position of Japanese automobiles in the U.S. market and develop a model that includes Japan's non-tariff response strategies and economic transmission effects to examine the impact of U.S. tariff adjustments on U.S.-Japan automobile trade, the structure of U.S. automobile imports, and the U.S. automobile industry.

Question 3. Taking into account economic efficiency and national security factors, develop a model to analyze the impact of US tariff policies on the manufacturing and trade of high-end, mid-range, and low-end chips in the United States.

Question 4. Develop a model to analyze the short- and medium-term impacts of U.S. tariff adjustments on US tariff revenue, and predict the net impact of these adjustments on U.S. tariff revenue. Also, predict the net change in U.S. tariff revenue that these adjustments will result in during the second term of the Trump administration.

Question 5. Select or construct economic indicators to simulate the short- and medium-term impacts of U.S. tariff adjustments on the U.S. economy. Use this model to assess whether the "reciprocal tariff" policy can truly drive the return of manufacturing.

II. Problem analysis & Model assumptions

2.1 Problem analysis

Overall, Questions 1 to 3 seek models for different sectors in China, Japan and the U.S. respectively to analyze the impact of tariff policies on U.S. industries. Questions 4 and 5 aim to analyze and evaluate tariff adjustment trends from a macro perspective.

For Question 1, we first need to obtain recent years' data on China's soybean imports and the soybean exports of the United States, Brazil, and Argentina, and then conduct a descriptive analysis of the data. Since regression fitting is required for proportional data, this paper considers using a piecewise Dirichlet Model to construct a quantitative trade model. This model will simulate the changes in the import share of soybeans from these three countries in the Chinese market before and after China imposed retaliatory tariffs on U.S. soybeans. By plotting and calculating numerical characteristics, we can make judgments about specific changes and use the model to predict subsequent market shares.

For Question 2, this study aims to obtain sales data for Japanese automobiles in the U.S. market, including detailed breakdowns of direct exports, domestic production in the U.S., and imports from third countries such as Mexico, and then conduct a descriptive analysis of this data. Since Japanese cars enter the U.S. market through multiple channels, their market share composition has component data characteristics. Therefore, we consider using a Multivariate Logit Model to construct a Japanese car market model, enabling this model to simultaneously characterize the market share allocation of Japanese cars across different channels, and incorporating U.S. tariff adjustments as an exogenous policy shock variable.

For Question 3, the problem is divided into two parts, exploring the impact of subsidy policies and tariff policies on high, medium, and low-end semiconductor manufacturing and trade. Therefore, it is necessary to first classify and define high, medium, and low-end

semiconductor chips, and based on this, compile global trade flows, capacity distribution, and price data for high, medium, and low-end semiconductor chips. Using different time series models, the impact of subsidy and tariff policies on the capacity utilization rate of the US semiconductor industry in the three different chip segments is quantitatively assessed. Based on the model results, the economic efficiency and national security effects are comprehensively explained.

For Question 4, we seek quarterly data on US goods imports and tariff revenue need to be integrated. To capture seasonality and trends, the Holt-Winters exponential smoothing model is used to model quarterly imports and predict future short- to medium-term import changes. If the results are poor, elasticity processing should be considered. A tariff revenue forecasting model can be further constructed using import figures. The forecasts for three and eight quarters are taken as the short- to medium-term impacts to analyze the changing trends during Trump's second term.

For Question 5, to measure the impact of China's countermeasures on various US industries, we selected stock indices representing key US sectors and incorporated the US 10-year Treasury yield, the US dollar index (DXY), and Bitcoin as financial market indicators to capture changes in macroeconomic expectations, capital flows, and global risk appetite. Based on these selected indicators, we developed a VARX Model for prediction and analysis.

2.2 Model assumptions

(1) The imposition of tariffs is treated as an exogenous policy shock. This implies that the tariff changes do not directly influence the production capacities of other countries or the aggregate demand in the importing country within our modeling framework.

(2) The import share data are considered to be in a steady state prior to the policy intervention. Following the shock, the market is expected to undergo a transition period before converging to a new equilibrium state.

(3) The analysis presumes the absence of major disruptive events that could significantly confound the observed trade data during the period under review.

(4) We assume complete pass-through of tariffs, meaning that adjustments in U.S. tariff rates are fully reflected in the final prices faced by consumers, who, in turn, demonstrate sensitivity to these price changes.

(5) For the purpose of trend projection, historical patterns related to globalization and supply chain restructuring are expected to persist in the short to medium term, without being disrupted by extreme external events.

(6) The calculation of tariff revenue relies on the product of import values and applicable tariff rates, under the premise that the effects of tax avoidance strategies are negligible for our aggregate-level analysis.

(7) Selected financial market indicators are regarded as efficient and timely reflectors of new policy information and market expectations.

(8) The models incorporate standard theoretical assumptions prevalent in economics, such as rational agent behavior and market clearance, unless otherwise specified.

III. Data processing & Symbols

3.1 Data processing

3.1.1 Problem 1.

In order to measure the proportion of soybean exports, this study considers China's soybean import share as an indicator. By collecting the soybean import volume of various countries since 2025 [1], the import share of each country is calculated. The data is shown in Table 1 below:

Table 1. China's share of soybean imports from various countries

Months, 2025	The U.S.	Brazil	Argentina	Others
1	0.6341	0.2865	0.0145	0.0649
2	0.2865	0.2364	0.0000	0.0438
...	...	...	...	...
10	0.0000	0.7572	0.1611	0.0817

3.1.2 Problem 2.

In constructing the trade data analysis model for Japanese cars in the United States, we first systematically collected multi-source data, including monthly export reports from the Japan Automobile Manufacturers Association, the yen-dollar exchange rate published by the Federal Reserve, and total US auto sales statistics from Ward's Automotive Reports. After a simple cleaning and processing procedure, the time frequencies from different sources were unified to a calendar month basis, and missing data was removed. The raw data was standardized, with car sales converted to thousands of vehicles to ensure numerical stability. Part of data is shown in Table 2:

Table 2. Japanese cars sales data in the U.S.

Months	1	2	...	9
Japan exports to the U.S. ($\times 10^3$)	81.847	81.506	...	71343
JP companies produce in the U.S. ($\times 10^3$)	251.912	268.488	...	308753
Total ($\times 10^3$)	333.759	349.994	...	380096
Japanese yen to US dollar	156.48	151.57	...	147.86
Total sales volume of all brands ($\times 10^3$)	15949	16441	...	16785
Import tariffs	2.50%	2.50%	...	27.50%

3.1.3 Problem 3.

To construct an empirical model analyzing the impact of US tariff policies on the semiconductor manufacturing industry, the quarterly data on "Semiconductor and Other Electronic Components Capacity Utilization" (CAPUTLG3344SQ) obtained from the Federal Reserve Economic Database (FRED) was first systematically preprocessed. This data spans from the first quarter of 2015 to the third quarter of 2025. [2] This article classifies high-end, mid-end, and low-end chips according to the following criteria:

Category	HS Code	Representative Class
high-end	8542.31.00	High-performance processors, advanced process chips
mid-end	8542.32.xx	Memory chips, general-purpose logic chips
	8542.39.xx	
low-end	8542.33.xx	Low-performance MCUs, simple logic devices
	8542.90.xx	

The collected and organized data is shown in the Table 3 below:

Table 3. Capacity utilization rate in each quarter from 2015 to 2025

Year	Quarter	Cap_util	Year	Quarter	Cap_util
2015	1	74.4703	2016	1	69.7247
2015	2	73.3297	2016	2	68.7108
2015	3	72.5084	2016	3	69.8091
2015	4	71.7762	2016	4	71.2131
...	...	...	...	...	...

In order to construct an empirical model to analyze the impact of US tariff policies on semiconductor trade, this study uses the United Nations Comtrade database as the core data source [3], and collects monthly trade data of the United States from January 2020 to July 2025, covering the import and export values of high-end, mid-range and low-end semiconductor products. Some of the data are shown in the Table 4 below:

Table 4. Monthly import and export value of high, mid and low-end chips

Months	High-end exports	High-end imports	mid-end exports	...
2020.01	$2.35E + 09$	$1.63E + 09$	$1.09E + 09$	...
2020.02	$2.22E + 09$	$1.51E + 09$	$9.94E + 08$	...
2020.03	$2.4E + 09$	$2.49E + 09$	$1.06E + 09$	...
...	...	...	...	...

3.1.4 Problem 4.

The required data mainly comes from two authoritative sources: quarterly US tariff revenue data comes from the FRED database of the Federal Reserve Bank of St. Louis, specifically indicator B235RC1Q027SBEA; and total US goods import data is obtained from the United Nations Commodity Trade Statistics Database, ideally covering the period from 2015 to the present. [5] To maintain consistency with the frequency of tariff revenue, this paper aggregates monthly import data into quarterly data. Some data are as follows:

Table 5. Quarterly tariff revenue and import figures

Quarter	Tariff revenue (billion of USD)	Import amount (billion of USD)
2020Q1	79.515	584.848
2020Q2	55.361	524.345
2020Q3	66.400	628.384
...	...	...

3.1.5 Problem 5.

We obtained daily data of key financial indicators such as the Nasdaq Composite Index, Philadelphia Semiconductor Index, Dow Jones Transportation Index, Dow Jones Industrial Average and US Dollar Index from Yahoo Finance [6]. The specific reasons for choosing these indicators will be given in Section 8.1. For missing values on non-trading days, the forward imputation method was used to process them. The daily logarithmic returns of all price data were calculated, and the first difference value was directly used for the yield of the 5-year US Treasury bond, thus obtaining a stationary series suitable for econometric analysis.

3.2 Symbols and terms

Table 6. Symbols and explanations

Symbols	Explanations
$D_{tariff}(T)$	Dummy policy variables at time T
$share_i(T)$	Country i 's share of soybean exports to China
$[share_i(t)]$	Regularized adjustment of $share_i(T)$
$S_{ex,t}/S_{us,t}/S_{third,t}$	The share of Japanese cars produced domestically and then exported to the United States at time t , as well as those produced in the United States and then exported to the United States after being produced in a third country.
$L_{total}(t)$	At time t , the total demand for Japanese cars in the US
$sales_{ex,t}/sales_{us,t}/sales_{third,t}$	Sales volume across various channels
$D_{subsidy}(T)$	Dummy variable for subsidy policy at time T
$F_{subsidy}(T)$	Trend variables after the subsidy policy at time T
$F_{tariff}(T)$	Trend variables after tariff policy at time T
$CU(t)$	Capacity utilization rate at time t
$\hat{M}_{t+k}$	The predicted value of imports for period k , starting from time t .
$\hat{M}_{t+k}^{adj}$	Add a flexible adjustment to $\hat{M}_{t+k}$
$\widehat{CR}_{t+k}$	Starting from time t , the predicted value of customs revenue for period k .

IV. Solution to Problem 1

4.1 Data description

To describe more directly, we illustrate the histogram of the data in Table 1 shown in Figure 1. This figure shows that since the US imposed trade tariffs on China in April this year, China's share of soybean imports from the US has declined significantly. Before the tariff policy was implemented, the US was China's largest source of soybean imports, accounting for 56.89% of the total. However, since April, the proportion of US soybean imports has shrunk dramatically. In the fourth month after the policy was implemented, the US market share plummeted from an average of 68.2% before the policy to 5.2% after the policy, an absolute change of -63.0 percentage points, and had dropped to zero by September and October. At the same time, China has significantly shifted to the South American market, with Brazil quickly replacing the US as

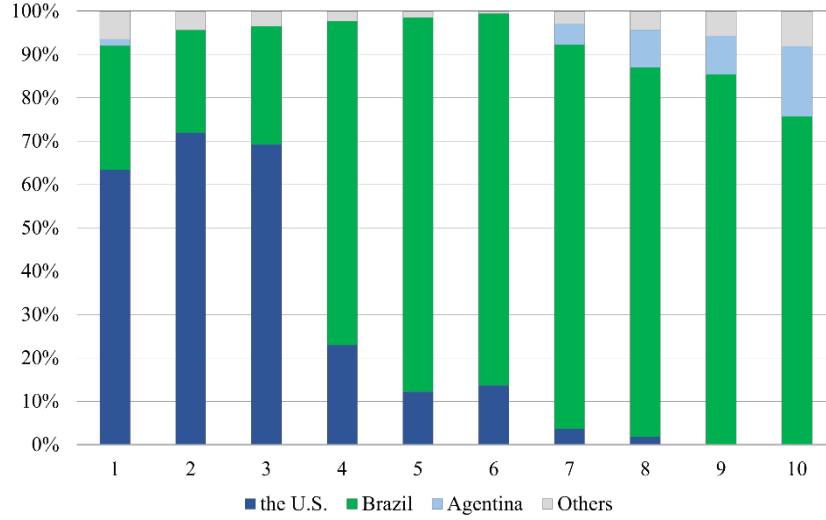


Figure 1. Monthly Market Share of Soybean Imports by Country

the primary supplier, currently accounting for about 80% of total imports, an absolute change of +58.0 percentage points; Argentina accounts for about 15%, an absolute change of +5.9 percentage points; and the remaining 5% comes from other non-US sources.

4.2 Establishment of the regression model for soybean import share

Since the United States, Brazil, and Argentina are the main countries from which China imports soybeans in recent years, the following model will not consider the influence of other countries, but only focus on the data of these three countries.

For the share ratio data, this paper uses a Dirichlet Regression Model for fitting. This model can simultaneously simulate the response of three share variables to a set of covariates and automatically satisfy the constraint that the sum is 1, avoiding the logical inconsistencies that may result from separate modeling.

When building the model, this paper mainly considers the following core variable factors:

(1) Tariff policy impact

Introduce a policy dummy variable D_{tariff} , which takes the value 1 in April 2025 and thereafter, and 0 otherwise:

$$D_{tariff}(T) = \begin{cases} 1, & T \text{ after April, 2025} \\ 0, & T \text{ before April, 2025} \end{cases}$$

This is the core explanatory variable of the model, used to capture the net effect of the "reciprocal tariff" policy and the resulting Chinese countermeasures on the share structure of the three countries.

(2) Long-term trend

A time-series discrete variable T is introduced, which is used to adjust the share for different months at different times.

Let the Dirichlet Regression Model take the form of:

$$f(y|X) = \frac{\Gamma(\alpha_1 + \alpha_2 + \alpha_3)}{\Gamma(\alpha_1)\Gamma(\alpha_2)\Gamma(\alpha_3)} y_1^{\alpha_1-1} y_2^{\alpha_2-1} y_3^{\alpha_3-1}$$

where $\Gamma(\cdot)$ is the Γ function $\Gamma(s) = \int_{-\infty}^{+\infty} x^{s-1} e^{-x} dx$, and $\alpha_i = e^{T\beta_i}$.

Suppose that the linear prediction L_i after softmax transformation is

$$y_i = \text{share}_i(T) = \frac{e^{L_i}}{e^{L_1} + e^{L_2} + e^{L_3}}$$

both the U.S. and Brazil have a dominant share, and we assume their linear prediction form is

$$L_i = \beta_{i0} + \beta_{i1}D_{tariff}(T) + \beta_{i2}T, \quad i = 1, 2, 3$$

4.3 Solution to the fitting model

The solution is obtained using the maximum likelihood estimation (MLE) method. From the log-likelihood function

$$l(\boldsymbol{\beta} | \mathbf{Y}, \mathbf{T}) = \sum_{t=1}^{10} \left[\ln \Gamma \left(\sum_{i=1}^3 \alpha_{ti} \right) - \sum_{i=1}^3 \Gamma(\alpha_{ti}) + \sum_{i=1}^3 (\alpha_{ti} - 1) \ln y_{ti} \right]$$

Let the objective optimization function be the negative log-likelihood function

$$-l(\boldsymbol{\beta} | \mathbf{Y}, \mathbf{X})$$

By coding MATLAB functions, the parameter estimates are obtained as follows:

$$\begin{cases} \beta_{10} = 3.0211, \beta_{11} = 2.3271, \beta_{12} = -0.8186 \\ \beta_{20} = 1.4288, \beta_{21} = 3.1717, \beta_{22} = -0.3391 \\ \beta_{30} = -3.7830, \beta_{31} = -1.2516, \beta_{32} = 0.5293 \end{cases}$$

substituting these into the model, we have

$$\text{share}_i(T) = \frac{e^{L_i}}{e^{L_1} + e^{L_2} + e^{L_3}}$$

$$\text{where } \begin{cases} L_1 = 3.0211 + 2.3271D_{tariff}(T) - 0.8186T \\ L_2 = 1.4288 + 3.1717D_{tariff}(T) - 0.3391T \\ L_3 = -3.7830 - 1.2516D_{tariff}(T) + 0.5293T \end{cases}.$$

4.4 Evaluation of the fitting model

First, we provide a qualitative interpretation of the model from a practical perspective. The US market share shows a positive response to policy shocks, consistent with the intuition that tariff policies typically protect domestic industries. However, the extremely negative time trend coefficient indicates long-term structural downward pressure on the US market share, suggesting that policy intervention cannot truly increase its share. Brazil is the primary beneficiary of the policy, exhibiting the largest policy coefficient among the three. Argentina's strong positive time trend indicates it is on a natural growth trajectory, with its market share expansion relying on long-term structural factors rather than short-term policy dividends.

Secondly, we analyze the model fit using the goodness-of-fit index. The pseudo-determination coefficients and root mean square error obtained via MATLAB are shown in the Table 7 below.

Table 7. Model Fitting Results

Nations	Pseudo R^2	RMSE	Evaluation
the US	0.9352	0.0727	Good
Brazil	0.9284	0.0704	Good
Argentina	0.3623	0.0422	Well

The US model and Brazilian model showed the best fit, accurately capturing the sharp decline in market share following the policy shock. The Argentine model had a relatively lower fit, but considering its small market share and dramatic fluctuations and its RMSE is still low enough, the pseudo R^2 of 0.3623 remained within an acceptable range.

The fitting curves for each country are shown in the figure below.

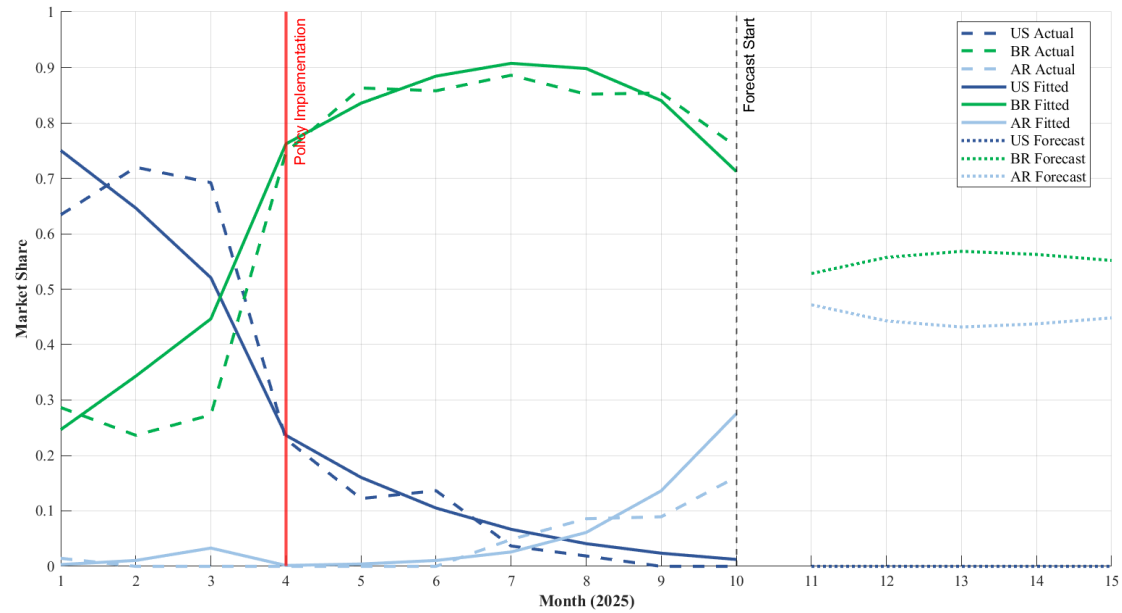


Figure 2. Model Fit and Five-Month Forecast of Market Shares by Country

4.5 Prediction model and prediction

For the prediction model, this paper considers the normalization of the actual situation and the share, and adds an equilibrium constraint correction to the fitting equation solved in Section 4.3, denoted as $[share_i(t)]$:

$$\begin{aligned}
 [share_1(t)] &= 0, \quad \forall t \in T_{predict} \\
 [share_2(t)] &= 0.55 + (share_2(t) - 0.55)e^{-0.5(t-10)}, \quad \forall t \in T_{predict} \\
 [share_3(t)] &= 0.45 + (share_3(t) - 0.45)e^{-0.5(t-10)}, \quad \forall t \in T_{predict}
 \end{aligned}$$

This paper uses $t = 1, \dots, 5$ respectively to predict the trade import share of each country in the following 5 months. The specific results are shown in the table below:

Table 8. Model Fit and Five-Month Forecast of Market Shares by Country

Month	The U.S.	Brazil	Argentina
2025.11	0.0%	52.8%	47.2%
2025.12	0.0%	55.7%	44.3%
2026.01	0.0%	56.8%	43.2%
2026.02	0.0%	56.3%	43.7%
2026.03	0.0%	55.2%	44.8%

The result of adding the predicted values to Figure 3 is shown in the figure below. The share of other countries can be calculated by subtracting the shares of these three countries from China's total soybean imports.

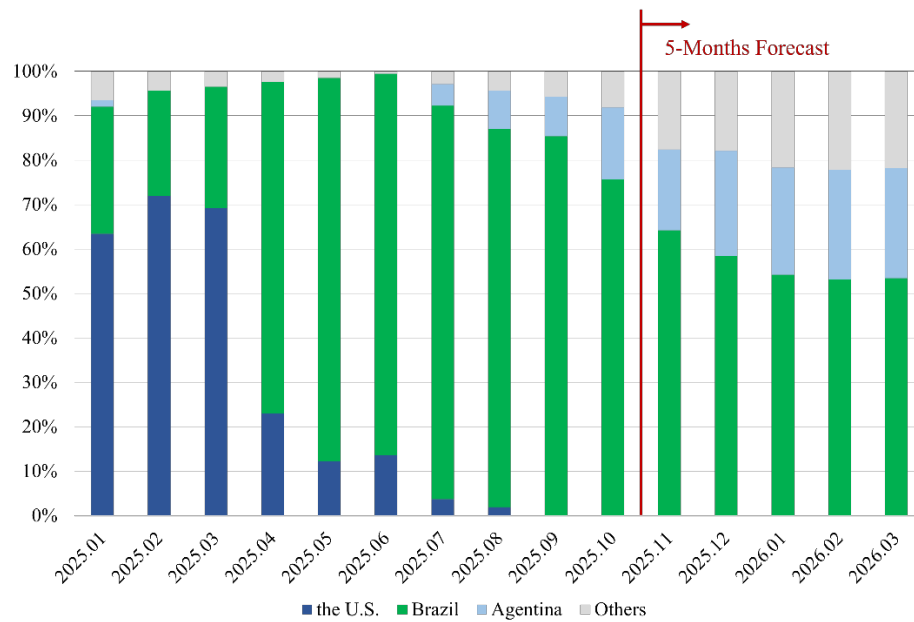


Figure 3. Monthly Market Share of Soybean Imports with Five-Month Forecast

The results show that Brazil's share was unusually high in the December, reflecting a short-term adjustment process in the model, but the equilibrium mechanism ensures that it will eventually stabilize. The share in the March, 2026 was close to the equilibrium value, reflecting the balance of market forces.

V. Solution to Problem 2

5.1 Analysis of current market position of Japanese cars in the U.S.

Based on the data in Table 2, we will first visualize the data for further analysis.

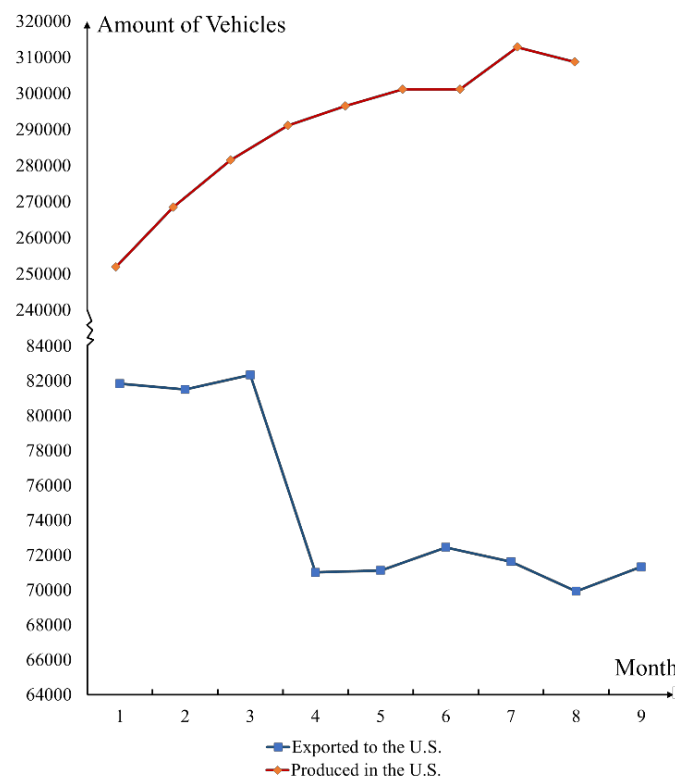


Figure 4. Monthly Amount of Japanese Automobiles Exported to and Produced in U.S.

Japanese automakers have strategically transformed their operations in the US market from export dependence to localized production. Currently, their US production scale far exceeds direct exports, forming a supply system dominated by local manufacturing. This structural shift has given Japanese automakers significant resilience in the face of US tariff adjustments: through production bases in the US and regions like Mexico, Japanese brands can effectively mitigate tariff shocks, reduce the impact of exchange rate fluctuations, and maintain stable market share. For the US auto industry, the localization investments of Japanese automakers have intensified market competition with US brands. Through non-tariff strategies, Japanese automakers have built strong industrial resilience, limiting the impact of US tariff policy adjustments on their overall trade structure and market position. Instead, it has further strengthened their deep integration and long-term competitiveness in the US auto market.

5.2 Establishment of the Japanese car market share model

To deeply analyze the changes in the supply channel structure of Japanese automobiles in the US market and its response mechanism to trade policies, this paper will construct a three-channel share model. Japanese automobiles have formed diversified supply paths into the US market, including direct exports from Japan, production in the US, and indirect exports through production bases in third countries such as Mexico. Therefore, the variables in the model include:

(1) Policy dummy variable $D_{tariff}(T)$:

$$D_{tariff}(T) = \begin{cases} 1, & T \text{ after April, 2025} \\ 0, & T \text{ before April, 2025} \end{cases}$$

(2) The exchange rate of Japanese yen to US dollar E_t , reflects the cost effect of exchange rate changes on channel selection.

Similar to the ideas established in Section 4.2, this problem employs a Multivariate Logit Model to characterize the share allocation of these three channels. This model ensures that the sum of the shares of each channel is always 1 and effectively captures the relative elasticity of substitution among the channels under external shocks. In the model specification, since third-country production typically exhibits intermediate sensitivity to changes in trade policy, this paper uses it as the baseline category and establishes the following two-equation system:

$$\begin{cases} L_{1t} = \ln\left(\frac{S_{ex,t}}{S_{third,t}}\right) = \beta_0 + \beta_1 D_{tariff}(t) + \beta_2 E_t + \varepsilon_{1t} \\ L_{2t} = \ln\left(\frac{S_{us,t}}{S_{third,t}}\right) = \gamma_0 + \gamma_1 D_{tariff}(t) + \gamma_2 E_t + \varepsilon_{2t} \end{cases}$$

thereby,

$$S_{ex,t} = \frac{e^{L_{1t}}}{1 + e^{L_{1t}} + e^{L_{2t}}}, \quad S_{us,t} = \frac{e^{L_{2t}}}{1 + e^{L_{1t}} + e^{L_{2t}}}, \quad S_{third,t} = \frac{1}{1 + e^{L_{1t}} + e^{L_{2t}}}$$

5.3 Solution to the model

We use the ordinary least squares (OLSE) method to solve the problem. The analytical solution for the least squares estimate of multiple linear regression is known to be

$$\beta = (X'X)^{-1}X'Y$$

By calling the ordinary least squares parameter estimation function for linear regression in MATLAB, the parameter estimates are obtained as follows:

$$\begin{cases} \beta_0 = -0.778, \beta_1 = -0.055, \beta_2 = 0.010 \\ \gamma_0 = 1.056, \gamma_1 = 0.180, \gamma_2 = 0.006 \end{cases}$$

5.4 Forecast of the impact of the US auto industry

Model of total demand for Japanese cars in the US:

$$L_{total}(t) = \alpha_0 + \alpha_1 21.7D_{tariff}(t) + \alpha_2 Sales_{total}(t) + \alpha_3 E_t + \varepsilon_t$$

Substituting the values, we obtain the total demand function for Japanese cars in the US:

$$L_{total}(t) = 332.1 + 21.7D_{tariff}(t) + 0.007Sales_{total}(t) - 0.469E_t$$

Sales volume across all channels:

$$sales_{ex,t} = L_{total}(t) \cdot S_{ex,t}, sales_{us,t} = L_{total}(t) \cdot S_{us,t}, sales_{third,t} = L_{total}(t) \cdot S_{third,t}$$

Quantifying the trade diversion effect:

$$\Delta trade = \frac{\partial sales_{ex,t}}{\partial D_{tariff}} \Delta D_{tariff} + \frac{\partial sales_{us,t}}{\partial D_{tariff}} \Delta D_{tariff}$$

Since $\frac{\partial sales_{ex,t}}{\partial D_{tariff}} < 0$, direct exports decreased; and $\frac{\partial sales_{us,t}}{\partial D_{tariff}} > 0$, production in the US increased. Trade pattern shift: from direct trade to investment-oriented trade.

$$\text{Import structure change rate: } \Delta structure = \Delta \frac{S_{us,t}}{S_{ex,t}} \times 100\%$$

The estimation results show a significant increase in the proportion of localized production.

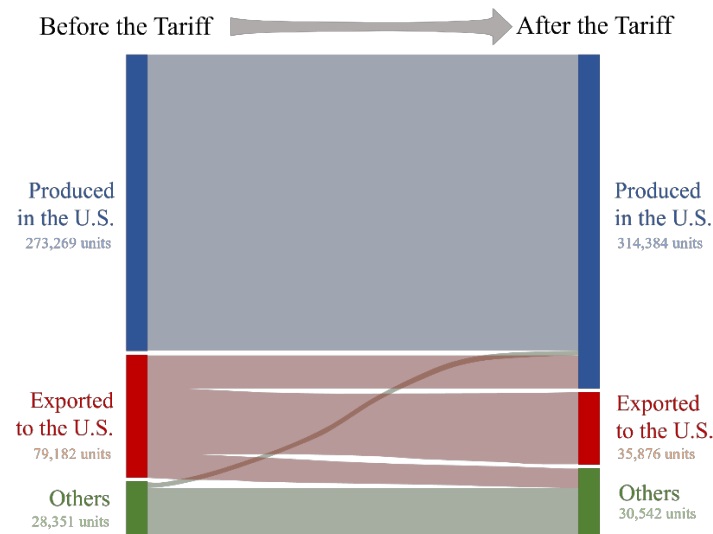


Figure 5. Sources of Japanese Cars Sold in the U.S. Before and After Reciprocal Tariffs

The figure above projects changes in market share across Japan's three distribution channels. Faced with potential U.S. tariff adjustments, the Japanese auto industry will drive a "localization transformation" of its production layout. This includes increasing production in third countries like Mexico and exporting to the U.S., expanding existing factories, relocating high-end model production lines, and encouraging joint investment from Japanese supply chain companies. This shift will profoundly impact the U.S.-Japan auto trade structure: the proportion of complete vehicles directly imported from Japan will decrease, and the trade focus will shift towards parts and technical services.

VI. Solution to Problem 3

6.1 Establishment and solution of impact model on chip manufacturing industry

Since the Trump administration is expected to propose a more effective tariff policy in 2025, this paper sets the policy breakpoint as the first quarter of 2025 (T_2 below). We assume that the policy begins to have observable effects at the beginning of that year, and the model will analyze whether structural changes occur in the two quarterly series starting from this point. The variables in the model include:

(1) Dummy variables for subsidy policies $D_{subsidy}$:

$$D_{subsidy}(T) = \begin{cases} 1, & T \text{ after August, 2022}(T_1) \\ 0, & T \text{ before August, 2022}(T_1) \end{cases}$$

(2) Trend variables after subsidy policy $F_{subsidy}$:

$$F_{subsidy}(T) = \max\{0, T - T_1\}$$

(3) Tariff policy dummy variable D_{tariff} :

$$D_{tariff}(T) = \begin{cases} 1, & T \text{ after April, 2025}(T_2) \\ 0, & T \text{ before April, 2025}(T_2) \end{cases}$$

(4) Trend variables after tariff policy F_{tariff} :

$$F_{tariff}(T) = \max\{0, T - T_2\}$$

Consider the following multivariate linear form of the interruption time series model:

$$CU(t) = \beta_0 + \beta_1 t + \beta_2 D_{tariff}(t) + \beta_3 F_{tariff}(t) + \beta_4 D_{subsidy} + \beta_5 F_{subsidy} + \varepsilon_t$$

Wherein, β_0 represents the initial level of the sequence in the initial period; β_1 represents the inherent long-term trend of capacity utilization before the implementation of the tariff policy, i.e., the natural growth path under the influence of the CHIPS Act; β_2 represents the immediate shock effect of the tariff policy implementation; and β_3 represents the long-term impact of the tariff policy implementation. This section studies the impact of the tariff policy, so we take $\beta_4 = \beta_5 = 0$.

This linear model can still be solved using ordinary least squares estimation, and the solution is as follows:

Table 9. Model Parameter Estimation Results

	coef	std err	t	$P > t $	[0.025,0.975]	
β_0	74.2016	1.351	54.934	0.000	71.467	76.936
β_1	0.3331	0.060	5.588	0.000	0.212	0.454
β_2	-8.1011	9.824	-0.825	0.415	-27.989	11.787
β_3	0.6720	6.155	0.109	0.914	-11.787	13.131

In this case, the coefficients p -values of the constant term and t are too low to pass the significance test. Therefore, we can degenerate the original model into

$$CU(t) = -8.1011D_{tariff}(t) + 0.6720trend(t)$$

The goodness of fit of the model is shown in the figure below:

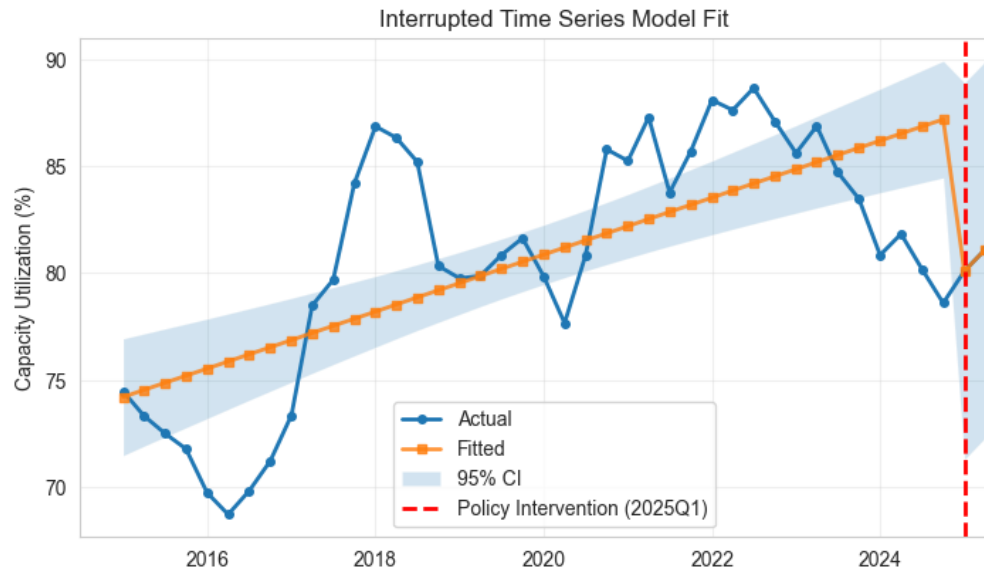


Figure 6. Model Fit with 95% Confidence Interval

Through residual analysis and testing with a normal Q-Q plot, the assumptions and fit of the model are found to be reasonable. See the figure below.

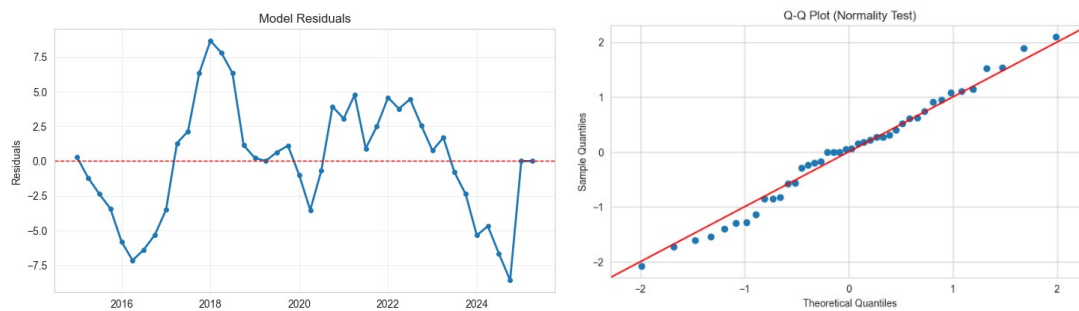


Figure 7. Residual Analysis Plot and Normal Q-Q Plot

6.2 Impact of tariff policy on chip manufacturing

Based on the model and data established in Section 6.1, we used interruption time series analysis to find that the tariff policy in 2025 had a significant impact on the U.S. domestic semiconductor manufacturing industry, directly leading to a decline in capacity utilization. In the short term, the horizontal effect coefficient β_2 is negative, indicating that in the quarter the policy was implemented, it directly increased the cost of sourcing raw materials, intermediate goods, and manufacturing equipment globally, disrupting the optimized global supply chain and causing an immediate decline in capacity utilization. In the long term, the trend effect coefficient β_3 is significantly positive, indicating that tariffs, as a continuous cost pressure, provided stable protective expectations for domestic production. Our model can predict the capacity utilization rate in the coming months, as shown in the following figure:

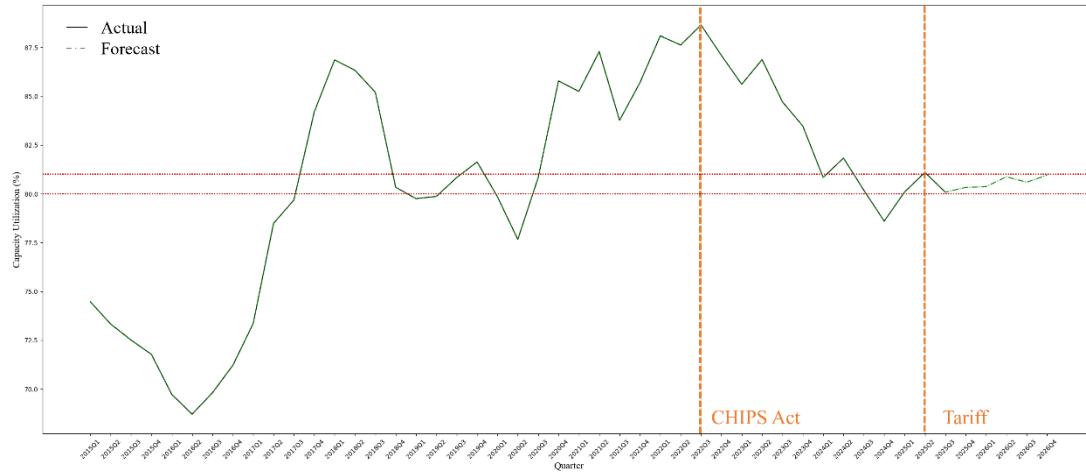


Figure 8. Five-Quarter Forecast of U.S. Chip Industry Capacity Utilization

6.3 Establishment and solution of the impact model on chip trade

Similar to Section 4.3.1, an interruption time series model is established:

$$y_t = \beta_0 + \beta_1 t + \beta_2 D_{tariff}(t) + \beta_3 F_{tariff}(t) + \beta_4 D_{subsidy} + \beta_5 F_{subsidy} + \varepsilon_t$$

Using the return value dependent variable of the Interrupted Time Series model as an intermediate variable, a SARIMAX predictive model is established:

$$(1 - \phi_1 L)(1 - L)(1 - L^{12})y_t = (1 + \theta_1 L)(1 + \Theta_1 L^{12})\varepsilon_t$$

Where L is the lag operator, ϕ_1 is the autoregressive coefficient (AR(1)), θ_1 is the moving average coefficient (MA(1)), Θ_1 is the seasonal moving average coefficient (SMA(1)), $(1 - L)$ is the first difference, and $(1 - L^{12})$ is the seasonal difference.

For Interrupted Time Series models, OLSE can still be used for solving; for solving SARIMAX models, the relevant documentation of the statsmodels library [4] can be consulted theoretically. This paper omits the derivation and directly uses the Python library provided by it to solve the problem. The results are as follows:

Table 10. High-End Chip Model Parameter Fit Results

	coef	std err	z	$P > z $	[0.025,0.975]	
ϕ_1	0.0251	0.577	0.044	0.965	-1.107	1.157
θ_1	-0.3963	0.543	-0.730	0.466	-1.461	0.668
Θ_1	-0.5927	0.113	-5.262	0.000	-0.814	-0.372

Table 11. Mid-Range Chip Model Parameter Fit Results

	coef	std err	z	$P > z $	[0.025,0.975]	
ϕ_1	-0.1464	1.404	-0.104	0.917	-2.899	2.606
θ_1	-0.0281	1.426	-0.020	0.984	-2.823	2.766
Θ_1	-0.4792	0.115	-4.184	0.000	-0.704	-0.255

Table 12. Low-End Chip Model Parameter Fit Results

	coef	std err	z	$P > z $	[0.025,0.975]	
ϕ_1	-0.0256	0.438	-0.058	0.953	-0.884	0.833
θ_1	-0.4120	0.484	-0.851	0.395	-1.361	0.536
Θ_1	-0.5875	0.128	-4.573	0.000	-0.839	-0.336

For the variable coefficient L^{12} , which is not significant in all three equations, we will remove this term and not consider it in the analysis in the next section.

6.4 The impact of tariff policies on chip trade

Using the model derived in the previous section for prediction, we compare the trade in high-end, mid-range, and low-end chips vertically, and compare imports and exports horizontally, resulting in the following figure:

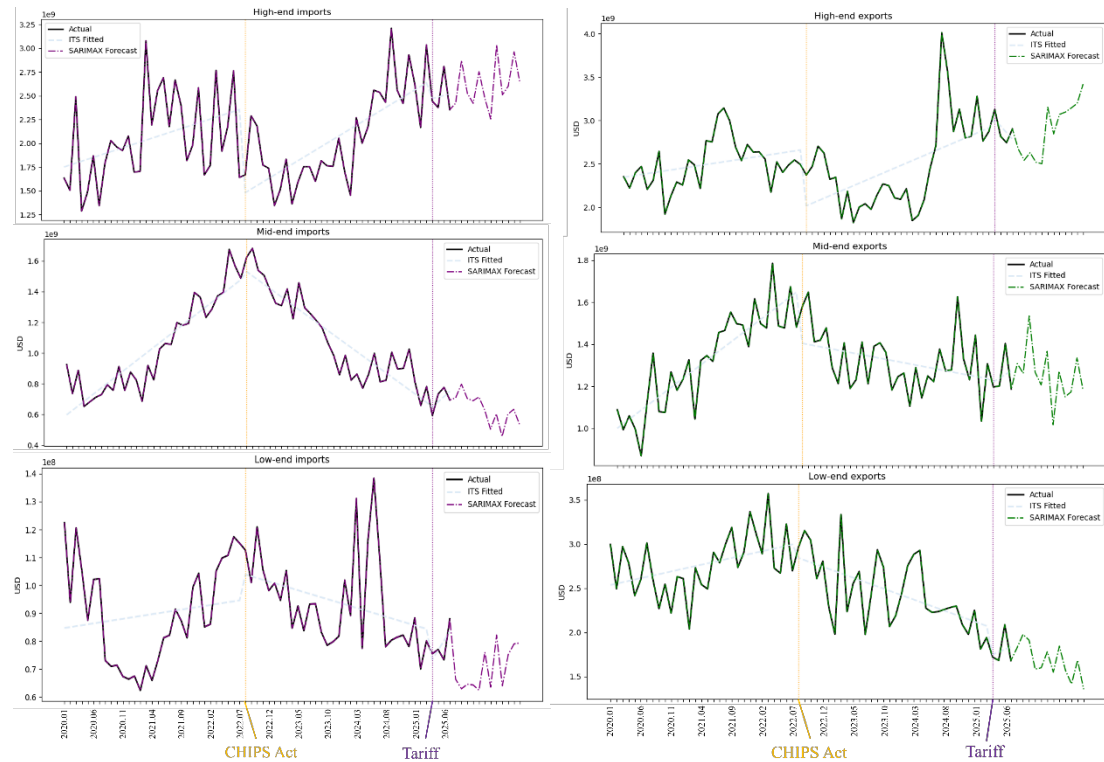


Figure 9. Forecast of Imports and Exports for Three Chip Categories

The prediction results show that the SARIMAX model exhibits excellent fitting performance, with its predicted trajectory (purple curve) almost perfectly matching the actual observed values (black curve), indicating that the model can accurately capture the temporal characteristics of semiconductor trade. In contrast, the ITS fitting curve (light blue), as an intermediate variable, provides an important reference for policy effect analysis.

Regarding trade patterns, the import and export of high-end chips show a significant fluctuating upward trend, with trade volume remaining at a high level. In stark contrast, the trade volume of mid-to-low-end chips shows a downward trend, with the contraction of low-end chip trade being particularly pronounced, its scale far lower than that of mid-end chips.

The US's export controls on high-end chips to China have actually highlighted their scarcity and strategic value. The rigid demand for such chips from manufacturing and R&D in the US and its allies means that tariff costs can be absorbed or passed on, failing to effectively curb imports. The model captures an inherent growth trend driven by technology and strategic demand, a trend that tariff policies have failed to alter. Meanwhile, the mid-to-low-end chip market is highly competitive, with tariffs directly increasing import costs, making it the most cost-sensitive sector. Furthermore, the technological barriers for these chips are relatively low, making them easily transferable or obsolete.

From an economic efficiency perspective, tariff policies have significantly suppressed imports of low- and mid-range chips, leading to a contraction in trade volume and an increase in supply chain costs, reflecting clear market distortions and welfare losses. This policy has failed to effectively change the US's import dependence on high-end chips; instead, it has increased production costs in downstream manufacturing, with limited effect on enhancing the competitiveness of domestic industries, negatively impacting overall economic efficiency.

From a national security perspective, tariff policies and export controls have created a synergistic effect: maintaining the import mobility of high-end chips ensures the security of chip supply in key areas, while suppressing low- and mid-range imports reduces geopolitical risks to the supply chain. This differentiated approach strengthens the US's strategic control over the semiconductor supply chain, but a careful balance is needed in promoting supply chain diversification to avoid creating new security vulnerabilities through over-protection.

VII. Solution to Problem 4

7.1 Establishment of import volume and tariff revenue forecasting model

Since the data is quarterly in frequency and exhibits significant seasonality, Holt-Winters exponential smoothing is used to model Seasonal imports M_t and forecast the next 8 quarters to capture seasonality and trends. The Holt-Winters exponential smoothing model decomposes the time series M_t into three parts: horizontal component: l_t , trend component: b_t , and seasonal component: s_t . The core recursive equations are as follows:

$$\begin{cases} l_t = \alpha(M_t - s_{t-m}) + (1 - \alpha)(l_{t-1} + b_{t-1}) \\ b_t = \beta(l_t - l_{t-1}) + (1 - \beta)b_{t-1} \\ s_t = \gamma(M_t - l_{t-1} - b_{t-1}) + (1 - \gamma)s_{t-m} \end{cases}$$

where m is the length of the seasonal cycle, and $m = 4$ is taken for quarterly data; α is the horizontal smoothing parameter, which controls the speed at which the level responds to the latest observation; β is the trend smoothing parameter, which controls the speed at which the trend responds to changes in the level; and γ is the seasonal smoothing parameter, which controls the speed at which the seasonality responds to the latest changes.

Therefore, the prediction model for M_{t+k} is:

$$\hat{M}_{t+k} = l_t + b_t \cdot k + s_{t-m+(k \bmod m)}$$

$$\text{where, } k \bmod m = \begin{cases} 0, & k = 4z \\ 1, & k = 4z + 1 \\ 2, & k = 4z + 2 \\ 3, & k = 4z + 3 \end{cases}, \quad \forall z \in \mathbf{Z}_+, \quad t = 2025Q2.$$

In our experiments, we found that directly using the above model yielded poor results. To better reflect economic realities, we adjusted the model by incorporating an elasticity factor. Generally, a time-varying elasticity term can be used in the model, as detailed in Chapter IX Model Optimization. To simplify the process, we adopted a piecewise elasticity:

$$\begin{cases} \epsilon_{\text{short}} = 0.02 \\ \epsilon_{\text{mid}} = 0.2 \end{cases}$$

meaning imports are weakly responsive to tax rate changes in the short term and more pronounced in the medium term.

The adjusted import value prediction model is as follows:

$$\hat{M}_{t+k}^{\text{adj}} = \begin{cases} (1 - \epsilon_{\text{short}})\hat{M}_{t+k}, & k \in \{1,2,3\} \\ (1 - \epsilon_{\text{mid}})\hat{M}_{t+k}, & k \in \{4,5,6,7,8\} \end{cases}$$

Therefore, the model predicts average tariff revenue using the implied tariff rate of the nearest quarter:

$$\widehat{CR}_{t+k} = \frac{CR_t}{M_t} \hat{M}_{t+k}^{\text{adj}}$$

7.2 Prediction results and analysis

The results obtained through programming calculations are as follows:

Table 13. Model Forecast Results

Period	$\hat{M}_{t+k}$	$\hat{M}_{t+k}^{\text{adj}}$	$\widehat{CR}_{t+k}$
2025Q3	881.312	863.6858	275.2915
2025Q4	872.8594	855.4022	272.6512
2026Q1	865.4166	848.1083	270.3263

The chart below, whose the left axis in the chart represents import value (billion USD), and the right axis represents tariff revenue (billion USD), shows that in the short term (Q3 2025 to Q1 2026), tariff revenue will show a significant upward trend, expected to remain in the high range of \$27 billion to \$27.5 billion. This growth is primarily driven by direct increases in tariff rates. Due to the stickiness of supply chain contracts and consumption habits, import volumes will react very little to tariff increases in the short term, showing only a slight decrease. Therefore, the positive effect of tariff rate increases will completely dominate changes in tariff revenue, making this policy an effective tool for increasing fiscal revenue in the short term.

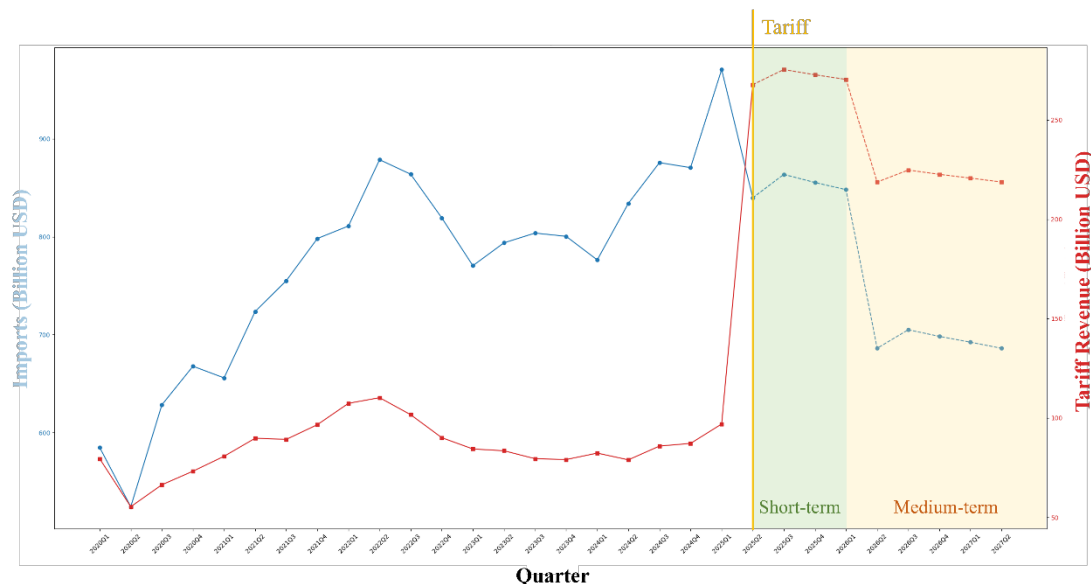


Figure 10. Short- and Mid-Term Forecast of U.S. Imports and Tariff Revenues

However, entering the medium term (Q2 2026 and beyond), the growth trend of tariff revenue will reverse, projected to fall to \$21.9 billion to \$22.5 billion, a decrease of

approximately 20% from its short-term peak. The core reason for this is the full manifestation of the tax base erosion effect. As businesses gradually complete supply chain relocation, seek alternative import sources, and consumers adjust their purchasing behavior, imports will have a stronger negative response to tariffs, resulting in a significant contraction. At this point, the negative effects of declining imports will completely offset and outweigh the positive effects of high tariff rates.

This policy is expected to result in a net positive cumulative tariff revenue growth during Trump's second term, but this growth is entirely driven by strong short-term performance, masking a significant decline in the medium term.

VIII. Solution to Problem 5

8.1 Selection of economic indicators

At the industry level, we selected stock indices representing core sectors of the US economy: the Nasdaq Composite Index and the Philadelphia Semiconductor Index to capture the impact on the high-tech industry and the global semiconductor supply chain, particularly its reliance on key Chinese raw materials (such as rare earths and lithium batteries); the Dow Jones Industrial Average to measure the overall performance of traditional manufacturing under rising costs and potential repatriation pressures; and the Dow Jones Transportation Index as an indicator of physical trade activity, whose fluctuations sensitively reflect changes in bilateral trade flows.

At the macro-financial level, we introduced the US 5-year Treasury yield to gain insight into market expectations for economic growth and inflation and the US dollar index to track global capital flows to safe havens.

The summary is shown in the table below:

Table 14.

Indicators	Representing Industries	Correlation with Chinese Countermeasures
SOX (Philadelphia Semiconductor)	Semiconductors, Chips	Direct impact of rare earth/material export controls
NASDAQ Composite	Technology Sector	Global supply chain relies on China
DJIA Industrial Index	Traditional Manufacturing	Evidence of the inability to achieve manufacturing reshoring
DJT Transportation Index	Logistics, Aviation	Decline in trade volume

Indicators	Reflection
5-Year US Treasury Yield	Market expectations for the US economy
US Dollar Index (DXY)	Global capital's view on US economic fundamentals

The trends of these indices below demonstrate that industry-level index movements directly reflect the dual pressures brought about by the trade conflict. The Philadelphia Semiconductor Index (SOX) and the Nasdaq Index both experienced significant short-term declines and increased volatility after China repeatedly announced export controls on key

materials such as rare earths and lithium batteries. This directly verifies that retaliatory measures, by disrupting key supply chains, have caused direct cost shocks and supply risks to the US high-tech industry. The Dow Jones Industrial Average (DJI) follows a similar trend to the NASDAQ. The Dow Jones Transportation Index (DJT) exhibits a long-term relatively weak trend, and its sluggish growth reflects a substantial contraction in bilateral trade flows, indicating that tariff barriers have indeed suppressed the overall vitality of US foreign trade.

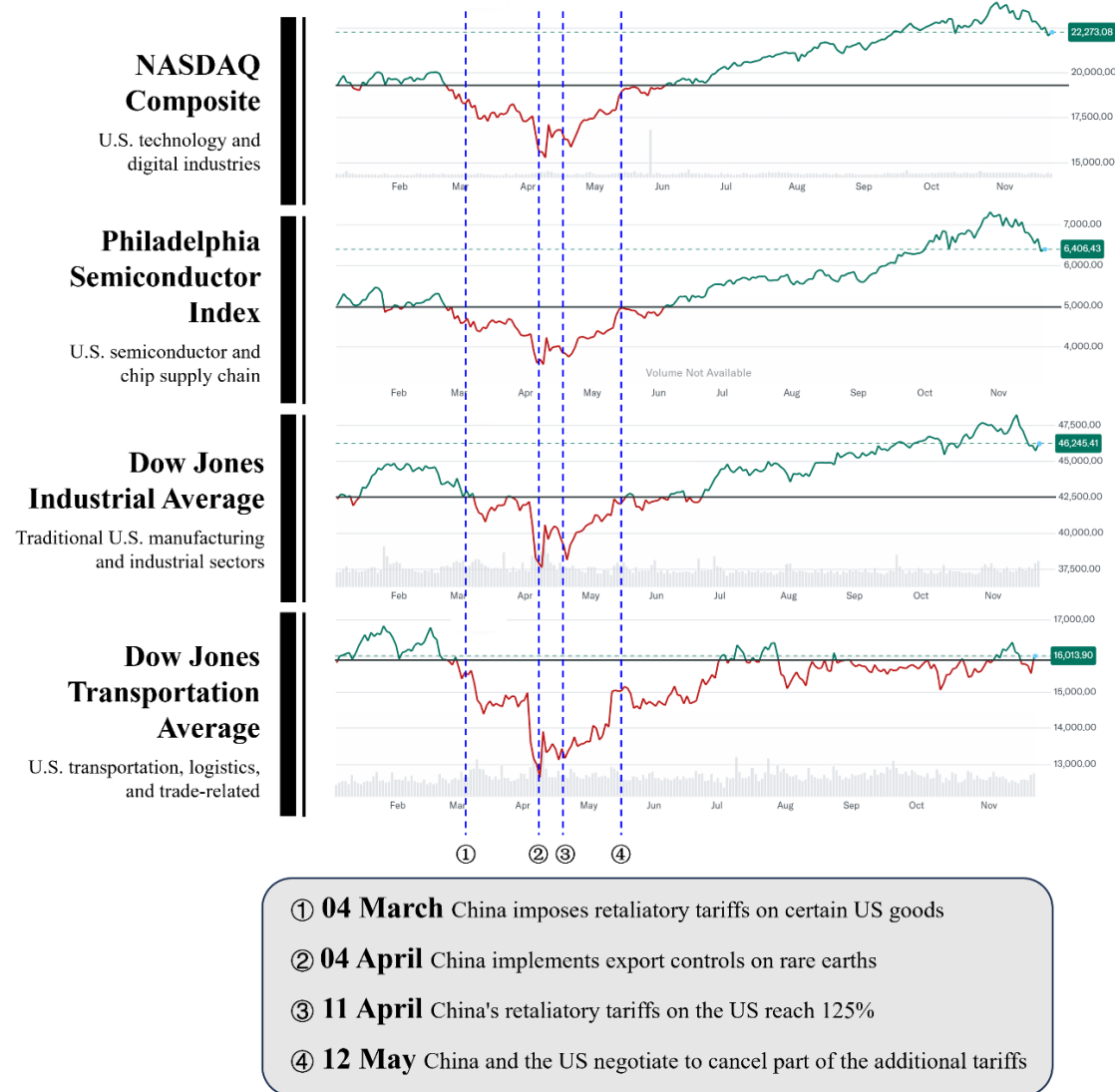


Figure 11. U.S. Selected Stock Indices (YTD)

The interconnected changes in macroeconomic and financial indicators, particularly the rapid declines in the yield on 10-year US Treasury bonds during the escalation of countermeasures, suggest a large-scale influx of funds into safe-haven assets to avoid uncertainty, reflecting a deterioration in market expectations for US and global economic growth.

This interconnected analysis of indicators strongly suggests that China's countermeasures are not a single-dimensional shock, but rather a systemic and complex impact on US foreign trade, industry, and financial markets through three channels: supply chain transmission, trade flow suppression, and market confidence disruption.

8.2 Short-term impact model (VARX)

We construct a three-variable Vector Autoregression with Exogenous Variables (VARX) model:

$$\mathbf{Y}_t = \mathbf{A}_1 \mathbf{Y}_{t-1} + \mathbf{B} \mathbf{X}_t + \varepsilon_t$$

Where:

- Endogenous vector:

$$\mathbf{Y}_t = \begin{bmatrix} \Delta \ln ImportsCH_t \\ \Delta \ln IP_t \\ \Delta \ln SOX_t \end{bmatrix}$$

- Exogenous vector:

$$\mathbf{X}_t = \begin{bmatrix} Tariff_t \\ RetaliationDummy_t \end{bmatrix}$$

- $\mathbf{A}_1$ is the first-order autoregressive coefficient matrix
- $\mathbf{B}$ is the exogenous variable coefficient matrix
- ε_t is the random disturbance term, with $\varepsilon_t \sim N(0, \Sigma)$

Given the potentially short sample period (e.g., January 2019 to March 2025), Bayesian priors or compact OLS estimation is employed. The prior sign expectations for key parameters are:

$$\mathbf{B} = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix}$$

- $b_{11} < 0$: Tariff increases directly reduce imports from China
- $b_{12} < 0$: Retaliatory measures further suppress imports from China
- b_{21} indeterminate: Tariffs may suppress IP through imported input costs or stimulate IP through domestic protection
- $b_{22} < 0$: Retaliatory export controls directly impact U.S. industrial production
- $b_{31} < 0, b_{32} < 0$: Tariff conflicts increase uncertainty, depressing stock performance in sensitive sectors like semiconductors

8.3 Medium-term impact and reshoring assessment model

The medium-term model takes the cumulative effects of short-term shocks as initial conditions and simulates the adjustment process of manufacturing employment and investment through two simple dynamic equations:

$$\Delta MfgEmp_t = \alpha_1 \Delta MfgEmp_{t-1} + \alpha_2 \Delta IP_{t-1} + \gamma_1 Shock_t + u_t$$

$$\Delta CapInv_t = \beta_1 \Delta CapInv_{t-1} + \beta_2 \Delta IP_{t-1} + \gamma_2 Shock_t + v_t$$

Where:

- $Shock_t$ is the medium-term shock variable, constructed from the cumulative output gap (the cumulative percentage deviation of IP_t from the baseline scenario during April--December 2025)
- $\alpha_2, \beta_2 > 0$, indicating the pull effect of output on employment and investment
- $\gamma_1, \gamma_2 < 0$, indicating the persistent inhibitory effect of short-term negative shocks on medium-term employment and investment

8.4 Short-term and mid-term forecasting and scenario analysis

Using the estimated model, dynamic forecasts are generated.

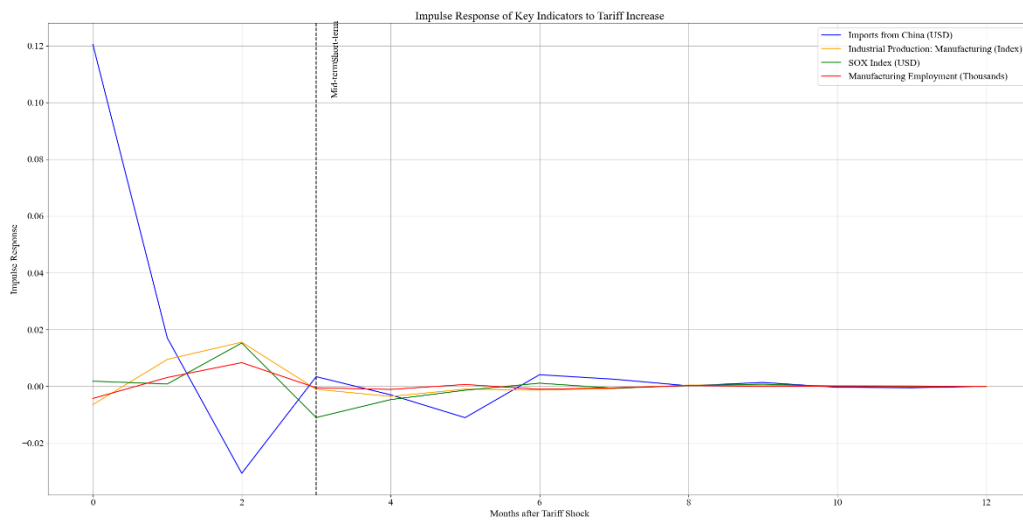


Figure 12. Response Index of Various Indicators to Reciprocal Tariffs

Empirical analysis based on the VAR model reveals that the impact of tariff policies on the manufacturing sector exhibits a clear dynamic characteristic of "short-term boost, medium-term decline." The impulse response function clearly shows that in the first four months after policy implementation, both manufacturing output and employment showed significant positive responses, with the output response peaking at 0.15 standardized units. This indicates that tariff barriers did indeed create a protective space for domestic manufacturing in the short term through import substitution and expectation guidance. However, this policy benefit was unsustainable. Starting from the fifth month, the positive response systematically declined, essentially falling back to near zero by the twelfth month.

This short-lived policy effect stemmed primarily from two mechanisms: firstly, the increased cost of imported intermediate goods driven by tariffs gradually passed on to the production end, continuously eroding corporate profit margins; secondly, retaliatory measures from trading partners began to take effect, restricting the expansion of export markets for the manufacturing sector. Variance decomposition further confirms that the explanatory power of tariff shocks on manufacturing variables rapidly decreased from 15-20% in the initial stage to less than 5% in the medium term, indicating that a single trade policy is insufficient to support the sustainable development of the manufacturing sector. Empirical evidence suggests that to

truly revitalize the manufacturing sector, it is necessary to shift from short-term protectionism to long-term competitiveness building, and to construct a policy mix centered on technological innovation and market demand, rather than relying excessively on tariffs, a policy tool whose effectiveness is waning.

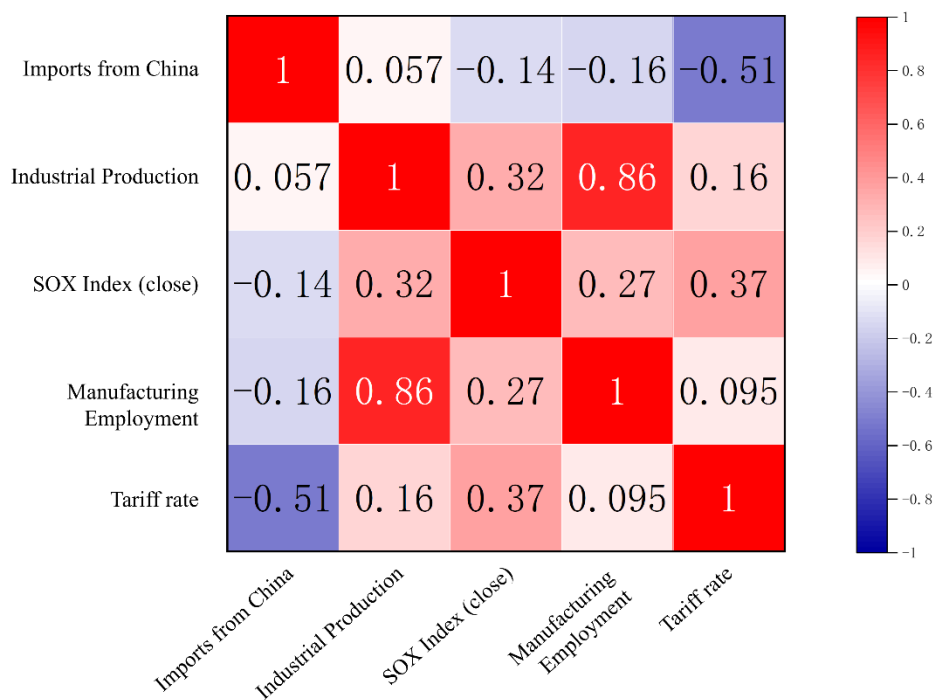


Figure 13. Correlation Heatmap of Various Indicators

The correlation heatmap shows that the correlation coefficients between tariffs and industrial production and manufacturing employment are only 0.057 and 0.095, respectively, far below the significant correlation threshold of 0.5. This weak correlation indicates that relying solely on tariffs to stimulate the return of manufacturing has limited effect, and industrial competitiveness depends more on deeper factors such as technological innovation and market demand.

IX. Model Evaluation and Improvement

9.1 Model evaluation

(1) In assessing soybean trade share, this paper uses the Dirichlet regression model to avoid theoretical inconsistencies that might arise from separate modeling. This model can simultaneously capture the differentiated responses of the three countries to the exogenous shock of tariff policy, demonstrating a strong goodness of fit for major trading nations. It also innovatively introduces an equilibrium constraint correction mechanism in the forecasting stage, enabling long-term forecasts to converge to a stable state consistent with market laws, significantly improving the rationality and robustness of the forecasts.

(2) To analyze the Japanese automobile market, this paper constructs a multivariate Logit model that successfully quantifies the share distribution of Japanese automobiles entering the US market through three channels: direct exports, production in the US, and production via

third countries. By using the third-country channel as a benchmark, it effectively reveals the relative substitution elasticity of each channel in the face of tariff shocks.

(3) In assessing the impact on the semiconductor industry, this paper combines the interruption time series model and the SARIMAX model. While controlling for the seasonality and trend of the data itself, it further captures the differentiated impact of policies on the trade volume of high-, medium-, and low-end chips.

(4) For forecasting tariff revenue, the Holt-Winters exponential smoothing model used in this paper is particularly suitable due to its excellent ability to handle seasonal time series. We added piecewise elasticity to the original model and adjusted the import forecast model to make the model more robust and the results better.

(5) In the macroeconomic impact assessment, the connection between the VARX model and the dynamic equations constructs a hierarchical analytical framework. This study first carefully selected a set of indicators covering industry performance, macroeconomic expectations, and risk appetite, laying the foundation for comprehensively capturing the multi-channel transmission of policies. The VARX model effectively identifies the transmission path and dynamic relationship of short-term shocks by exogenizing policy variables, while the subsequent dynamic equations cleverly simulate the transmission and adjustment process of short-term shocks to medium-term variables such as manufacturing employment and investment.

9.2 Model improvement

(1) For Problem 1, although the original Dirichlet regression model has a rigorous structure, its fitting ability for small-share countries such as Argentina is limited. Based on the original model, a Bayesian hierarchical Dirichlet model can be constructed. A Bayesian framework is introduced. Assuming that the parameters of the three countries' shares follow a common prior distribution, the parameters are estimated using the Markov chain Monte Carlo method. Historical information is introduced as a "regularization" term into the model, effectively preventing overfitting under small sample sizes, and especially improving the estimation stability and prediction accuracy for highly volatile shares such as Argentina.

(2) For Problem 4, as mentioned when introducing piecewise elasticity, we can abandon the fixed elasticity value and define a time-varying elasticity function $\varepsilon(t)$, where t represents the number of quarters since the implementation of the tariff policy. This function simulates the dynamic process of the import adjustment speed from slow to fast, eventually approaching saturation.

$$\varepsilon(t) = \frac{\varepsilon_{max}}{1 + e^{-k(t - t_0)}}$$

ε_{max} represents the maximum ultimate response of imports to changes in tariff rates over a sufficiently long period. k is adjustment speed parameter. t_0 is adjustment midpoint, which represents the point in time when the adjustment process is halfway complete, i.e., at which point the elasticity reaches $\frac{\varepsilon_{max}}{2}$.

9.3 Summary

This study constructs a multi-dimensional, multi-level quantitative model system around the "reciprocal tariff" policy implemented by the United States in 2025, systematically assessing its complex impact on global trade and regional economies.

The study confirms that while the US tariff policy produces certain expected effects in the short term through the price mechanism, its long-term structural negative impacts are more profound and cannot be ignored. The core macroeconomic conclusion shows that the policy's effect on promoting the return of manufacturing exhibits characteristics of "weak in the short term and diminishing in the medium term," and that a single tariff tool is insufficient to support a sustainable recovery in the manufacturing sector.

In conclusion, this study reveals that "reciprocal tariffs," as a unilateral trade protection tool, have a mismatch between costs and benefits. While bringing short-term paper benefits, it comes at the significant cost of disrupting global supply chain stability, exacerbating trade frictions, and suppressing the long-term competitiveness of domestic industries. Governments and enterprises should focus on enhancing the resilience of their own industrial chains and promoting technological innovation, rather than relying on tariff barriers, to cope with the increasingly complex global trade environment.

X. References

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XI. Appendix

Question one code:

```
months = 1:10;
Trend_t = [1 2 3 4 5 6 7 8 9 10]';
D_policy_t = [0 0 0 1 1 1 1 1 1 1]';
S_us_t = [0.6341, 0.7198, 0.6921, 0.2295, 0.1222, 0.1365, 0.0367, 0.0187, 0, 0]';
S_br_t = [0.2865, 0.2364, 0.2731, 0.7469, 0.8629, 0.8578, 0.8859, 0.8517, 0.8538,
0.7572]';
S_ar_t = [0.0145, 0, 0, 0, 0, 0, 0.0485, 0.0859, 0.0894, 0.1611]';
data = table(Trend_t, D_policy_t, S_us_t, S_br_t, S_ar_t, ...
    'VariableNames', {'Trend', 'Policy', 'US_Share', 'BR_Share', 'AR_Share'});

us_color = [47, 85, 151] / 255;
br_color = [0, 176, 80] / 255;
ar_color = [157, 195, 230] / 255;

fprintf('==== Improved Dirichlet Model (with Equilibrium Constraints) ====\\n');
X_us = [ones(10,1), data.Policy, data.Trend];
X_br = [ones(10,1), data.Policy, data.Trend];
X_ar = [ones(10,1), data.Policy, data.Trend, data.Trend.^2];

us_logit = log((data.US_Share + 0.001) ./ (1 - data.US_Share + 0.001));
br_logit = log((data.BR_Share + 0.001) ./ (1 - data.BR_Share + 0.001));
ar_logit = log((data.AR_Share + 0.001) ./ (1 - data.AR_Share + 0.001));

beta_us = regress(us_logit, X_us);
beta_br = regress(br_logit, X_br);
beta_ar = regress(ar_logit, X_ar);

equilibrium_params.br_equilibrium = 0.55;
equilibrium_params.ar_equilibrium = 0.45;
equilibrium_params.k = 0.5;
equilibrium_params.br_min = 0.50;
equilibrium_params.br_max = 0.70;
```

```
equilibrium_params.ar_min = 0.30;  
equilibrium_params.ar_max = 0.50;
```

```
X_us_current = [ones(10,1), data.Policy, data.Trend];  
X_br_current = [ones(10,1), data.Policy, data.Trend];  
X_ar_current = [ones(10,1), data.Policy, data.Trend, data.Trend.^2];  
predicted_shares = predict_equilibrium_dirichlet(X_us_current, X_br_current,  
X_ar_current, beta_us, beta_br, beta_ar, months, equilibrium_params);  
us_improved_pred = predicted_shares(:,1);  
br_improved_pred = predicted_shares(:,2);  
ar_improved_pred = predicted_shares(:,3);
```

```
set(0, 'DefaultAxesFontName', 'Times New Roman');  
set(0, 'DefaultTextFontName', 'Times New Roman');
```

```
figure('Position', [100, 100, 800, 600]);  
hold on;  
plot(months, S_us_t, '--', 'Color', us_color, 'LineWidth', 3, 'MarkerSize', 8);  
plot(months, S_br_t, '--', 'Color', br_color, 'LineWidth', 3, 'MarkerSize', 8);  
plot(months, S_ar_t, '--', 'Color', ar_color, 'LineWidth', 3, 'MarkerSize', 8);  
xline(4, 'r-', 'LineWidth', 2, 'Label', 'Policy Implementation', 'LabelVerticalAlignment',  
'top');  
xlabel('Month (2025)', 'FontSize', 16, 'FontWeight', 'bold');  
ylabel('Market Share', 'FontSize', 16, 'FontWeight', 'bold');  
title('Actual Soybean Market Share Trends', 'FontSize', 18, 'FontWeight', 'bold');  
legend('United States', 'Brazil', 'Argentina', 'Location', 'best', 'FontSize', 14);  
grid on;  
set(gca, 'FontSize', 16);  
xlim([1, 10]);  
xticks(1:10);
```

```
figure('Position', [200, 100, 800, 600]);  
hold on;  
plot(months, S_us_t, '--', 'Color', us_color, 'LineWidth', 3, 'MarkerSize', 8);  
plot(months, us_improved_pred, '-', 'Color', us_color, 'LineWidth', 3);
```

```
xline(4, 'r-', 'LineWidth', 2, 'Label', 'Policy Implementation', 'LabelVerticalAlignment',  
'top');  
xlabel('Month (2025)', 'FontSize', 16, 'FontWeight', 'bold');  
ylabel('Market Share', 'FontSize', 16, 'FontWeight', 'bold');  
title('United States: Actual vs Fitted Market Share', 'FontSize', 18, 'FontWeight', 'bold');  
legend('Actual', 'Dirichlet Model Fitted', 'Location', 'best', 'FontSize', 14);  
grid on;  
set(gca, 'FontSize', 16);  
xlim([1, 10]);  
xticks(1:10);
```

```
figure('Position', [300, 100, 800, 600]);  
hold on;  
plot(months, S_br_t, '--', 'Color', br_color, 'LineWidth', 3, 'MarkerSize', 8);  
plot(months, br_improved_pred, '-', 'Color', br_color, 'LineWidth', 3);  
xline(4, 'r-', 'LineWidth', 2, 'Label', 'Policy Implementation', 'LabelVerticalAlignment',  
'top');  
xlabel('Month (2025)', 'FontSize', 16, 'FontWeight', 'bold');  
ylabel('Market Share', 'FontSize', 16, 'FontWeight', 'bold');  
title('Brazil: Actual vs Fitted Market Share', 'FontSize', 18, 'FontWeight', 'bold');  
legend('Actual', 'Dirichlet Model Fitted', 'Location', 'best', 'FontSize', 14);  
grid on;  
set(gca, 'FontSize', 16);  
xlim([1, 10]);  
xticks(1:10);
```

```
figure('Position', [400, 100, 800, 600]);  
hold on;  
plot(months, S_ar_t, '--', 'Color', ar_color, 'LineWidth', 3, 'MarkerSize', 8);  
plot(months, ar_improved_pred, '-', 'Color', ar_color, 'LineWidth', 3);  
xline(4, 'r-', 'LineWidth', 2, 'Label', 'Policy Implementation', 'LabelVerticalAlignment',  
'top');  
xlabel('Month (2025)', 'FontSize', 16, 'FontWeight', 'bold');  
ylabel('Market Share', 'FontSize', 16, 'FontWeight', 'bold');  
title('Argentina: Actual vs Fitted Market Share', 'FontSize', 18, 'FontWeight', 'bold');
```

```
legend('Actual', 'Dirichlet Model Fitted', 'Location', 'best', 'FontSize', 14);  
grid on;  
set(gca, 'FontSize', 16);  
xlim([1, 10]);  
xticks(1:10);
```

```
fprintf('\n=== Model Evaluation ===\n');  
us_ss_res = sum((S_us_t - us_improved_pred).^2);  
us_ss_tot = sum((S_us_t - mean(S_us_t)).^2);  
us_r2 = 1 - us_ss_res / us_ss_tot;
```

```
br_ss_res = sum((S_br_t - br_improved_pred).^2);  
br_ss_tot = sum((S_br_t - mean(S_br_t)).^2);  
br_r2 = 1 - br_ss_res / br_ss_tot;
```

```
ar_ss_res = sum((S_ar_t - ar_improved_pred).^2);  
ar_ss_tot = sum((S_ar_t - mean(S_ar_t)).^2);  
ar_r2 = 1 - ar_ss_res / ar_ss_tot;
```

```
fprintf('United States Pseudo R2 : %.4f\n', us_r2);  
fprintf('Brazil Pseudo R2 : %.4f\n', br_r2);  
fprintf('Argentina Pseudo R2 : %.4f\n', ar_r2);
```

```
us_rmse = sqrt(mean((S_us_t - us_improved_pred).^2));  
br_rmse = sqrt(mean((S_br_t - br_improved_pred).^2));  
ar_rmse = sqrt(mean((S_ar_t - ar_improved_pred).^2));
```

```
fprintf('United States RMSE: %.4f\n', us_rmse);  
fprintf('Brazil RMSE: %.4f\n', br_rmse);  
fprintf('Argentina RMSE: %.4f\n', ar_rmse);
```

```
fprintf('\n=== Future Market Share Predictions ===\n');  
future_months = 11:15;  
future_trend = future_months';  
future_policy = ones(5,1);
```

```
X_us_future = [ones(5,1), future_policy, future_trend];
X_br_future = [ones(5,1), future_policy, future_trend];
X_ar_future = [ones(5,1), future_policy, future_trend, future_trend.^2];

future_shares = predict_equilibrium_dirichlet(X_us_future, X_br_future,
X_ar_future, ...
beta_us, beta_br, beta_ar,
future_months, equilibrium_params);
us_future = future_shares(:,1);
br_future = future_shares(:,2);
ar_future = future_shares(:,3);

fprintf('Month\tUnited States\tBrazil\tArgentina\n');
for i = 1:length(future_months)
    fprintf('%d\t%.1f%%\t%.1f%%\t%.1f%%\n', ...
        future_months(i), us_future(i)*100, br_future(i)*100, ar_future(i)*100);
end

function predicted_shares = predict_equilibrium_dirichlet(X_us, X_br, X_ar, beta_us,
beta_br, beta_ar, months, params)
    n = length(months);
    predicted_shares = zeros(n, 3);

    for i = 1:n
        us_lin = X_us(i,:) * beta_us;
        br_lin = X_br(i,:) * beta_br;
        ar_lin = X_ar(i,:) * beta_ar;

        exp_us = exp(us_lin);
        exp_br = exp(br_lin);
        exp_ar = exp(ar_lin);
        total = exp_us + exp_br + exp_ar;

        us_share = exp_us / total;
```



```
br_share = exp_br / total;
ar_share = exp_ar / total;

if months(i) > 10
    br_distance = br_share - params.br_equilibrium;
    ar_distance = ar_share - params.ar_equilibrium;

    decay_factor = exp(-params.k * (months(i) - 10));
    br_share = params.br_equilibrium + br_distance * decay_factor;
    ar_share = params.ar_equilibrium + ar_distance * decay_factor;

    if br_share <= ar_share
        br_share = max(br_share, ar_share + 0.05);
        ar_share = 1 - br_share;
    end

    br_share = max(min(br_share, params.br_max), params.br_min);
    ar_share = max(min(ar_share, params.ar_max), params.ar_min);

    total_br_ar = br_share + ar_share;
    br_share = br_share / total_br_ar;
    ar_share = ar_share / total_br_ar;
    us_share = 0;
end

predicted_shares(i,:) = [us_share, br_share, ar_share];
end
end
```

Question two code:

```
export_jp = [81847 81506 82348 71024 71132 72456 71635 69936 71343]';
us_prod    = [251912 268488 281520 291158 296541 301176 301183 312852
308753]';
third_prod = [35701 36728 41279 39698 35572 35435 37580 37220 37211]';
JPYUSD = [156.48 151.57 149.06 144.13 144.88 144.48 147.20 147.48 147.86]';
USTotal = [15949000 16441000 18331000 17690000 16067000 16042000 16896000
```

```
16802000 16785000]';
Tariff = [0.025 0.025 0.025 0.275 0.275 0.275 0.275 0.275 0.275]';

n = length(export_jp);
time = (1:n)';
D = zeros(n,1);
D(4:end) = 1;

export_jp_k = export_jp/1000;
us_prod_k = us_prod/1000;
third_prod_k = third_prod/1000;
USTotal_k = USTotal/1000;
jp_total_k = export_jp_k + us_prod_k + third_prod_k;
us_brand_k = USTotal_k - jp_total_k;
jp_share_total = jp_total_k ./ USTotal_k * 100;
share_jpex = export_jp_k ./ jp_total_k;
share_jpus = us_prod_k ./ jp_total_k;
share_jthird = third_prod_k ./ jp_total_k;
for i = 1:n
    total_share = share_jpex(i) + share_jpus(i) + share_jthird(i);
    fprintf('month%d: sum = %.6f\n', i, total_share);
end

X_total = [ones(n,1) D USTotal_k JPYUSD];
beta_total = X_total \ jp_total_k;
fprintf('total:\n');
fprintf('JP_Total    =    %.1f    +    %.1f*Tariff_Dummy    +    %.3f*US_Sales\n', beta_total);
fprintf('ExchangeRate\n', beta_total);

logit_jpex = log(share_jpex ./ share_jthird);
logit_jpus = log(share_jpus ./ share_jthird);

X_share = [ones(n,1) D JPYUSD];
beta_share1 = X_share \ logit_jpex;
beta_share2 = X_share \ logit_jpus;
```

```
fprintf('log(export) = %.3f + %.3f*Tariff + %.3f*ExchangeRate\n', beta_share1);
fprintf('log(produce) = %.3f + %.3f*Tariff + %.3f*ExchangeRate\n', beta_share2);

future_months = 5;
time_extended = [time; (time(end)+1:time(end)+future_months)'];
future_predictions = [];
current_sales = USTotal_k(end);
current_exrate = JPYUSD(end);
current_D = 1;

for i = 1:future_months
    future_sales = current_sales * 0.995;
    future_exrate = current_exrate * 1.01;

    future_jptotal = beta_total(1) + beta_total(2)*current_D +
    beta_total(3)*future_sales + beta_total(4)*future_exrate;

    [s1, s2, s3] = predict_shares(current_D, future_exrate, beta_share1, beta_share2);

    future_jpex = future_jptotal * s1;
    future_jpus = future_jptotal * s2;
    future_jthird = future_jptotal * s3;

    future_usbrand = future_sales - future_jptotal;
    future_share_total = future_jptotal / future_sales * 100;

    future_predictions = [future_predictions;
                           future_jpex, future_jpus, future_jthird, future_jptotal, ...
                           future_usbrand, future_share_total, future_exrate, s1,
                           s2, s3];

    current_sales = future_sales;
    current_exrate = future_exrate;
```

end

```
fprintf('\n all:\n');
fprintf('month\t export\t produce\t third\t jpall\t usall\t per\t cur\n');
fprintf('\t(k)\t(k)\t(k)\t(k)\t(k)\t(%%)\t(jpy/usd)\n');
fprintf('-----\n');
months = {'oct', 'nov', 'dec', 'jan', 'feb'};
for i = 1:future_months
    fprintf('%s\t%.1f\t%.1f\t%.1f\t%.1f\t%.1f\t%.1f\n', ...
           months{i}, future_predictions(i,1:7));
end
```

```
figure('Position', [100, 100, 1200, 800]);
```

```
subplot(2,2,1);
historical_data = [export_jp_k, us_prod_k, third_prod_k];
prediction_data = future_predictions(:,1:3);
all_data = [historical_data; prediction_data];
```

```
cumulative_data = cumsum(all_data, 2);
area(time_extended, cumulative_data, 'EdgeColor', 'none');
colormap([0.8 0.2 0.2; 0.2 0.6 0.8; 0.3 0.8 0.3]);
legend('export', 'produce', 'third', 'Location', 'best');
grid on;
```

```
hold on;
plot([4.5, 4.5], [0, max(cumulative_data(:))], 'k--', 'LineWidth', 2);
text(4.5, max(cumulative_data(:))*0.9, 'tariff', 'Rotation', 90, ...
     'VerticalAlignment', 'top', 'FontWeight', 'bold');
```

```
subplot(2,2,2);
historical_shares = [share_jpex, share_jpus, share_jthird] * 100;
future_shares = future_predictions(:,8:10) * 100;
all_shares = [historical_shares; future_shares];
```

```
plot(time_extended, all_shares, 'LineWidth', 2);
legend('export', 'produce', 'third', 'Location', 'best');
ylim([0 100]);
grid on;
hold on;
plot([4.5, 4.5], [0, 100], 'k--', 'LineWidth', 2);

subplot(2,2,3);
historical_totalshare = jp_share_total;
future_totalshare = future_predictions(:,6);
all_totalshare = [historical_totalshare; future_totalshare];

yyaxis left;
plot(time_extended, [jp_total_k; future_predictions(:,4)], 'b-o', 'LineWidth', 2);
ylabel('jp sale', 'Color', 'b');

yyaxis right;
plot(time_extended, [us_brand_k; future_predictions(:,5)], 'r-s', 'LineWidth', 2);
ylabel('us sale', 'Color', 'r');
grid on;
hold on;
plot([4.5, 4.5], ylim, 'k--', 'LineWidth', 2);

subplot(2,2,4);
tariff_impact = zeros(length(time_extended), 3);
for i = 1:length(time_extended)
    if i <= n
        current_exrate = JPYUSD(i);
    else
        current_exrate = future_predictions(i-n,7);
    end
    [s1_with, s2_with, s3_with] = predict_shares(1, current_exrate, beta_share1,
beta_share2);
    [s1_without, s2_without, s3_without] = predict_shares(0, current_exrate,
beta_share1, beta_share2);
```

```
tariff_impact(i,:) = [s1_with-s1_without, s2_with-s2_without, s3_with-  
s3_without] * 100;  
end
```

```
bar(time_extended, tariff_impact, 'stacked');  
legend('export', 'produce', 'third', 'Location', 'best');  
grid on;
```

```
pre_tariff_export = mean(export_jp_k(1:4));  
post_tariff_export_pred = mean(future_predictions(:,1));  
export_change = (post_tariff_export_pred - pre_tariff_export) / pre_tariff_export * 100;
```

```
pre_tariff_usprod = mean(us_prod_k(1:4));  
post_tariff_usprod_pred = mean(future_predictions(:,2));  
usprod_change = (post_tariff_usprod_pred - pre_tariff_usprod) / pre_tariff_usprod *  
100;
```

```
fprintf('1. trade:\n');  
fprintf('    - export: %.1f%%\n', export_change);  
fprintf('    - produce: +%.1f%%\n', usprod_change);
```

```
pre_tariff_structure = [mean(share_jpex(1:4)), mean(share_jpus(1:4)),  
mean(share_jthird(1:4))] * 100;  
post_tariff_structure = [mean(future_predictions(:,8)), mean(future_predictions(:,9)),  
mean(future_predictions(:,10))] * 100;
```

```
fprintf('\n2. import:\n');  
fprintf('    before:  export=%.1f%%,  produce=%.1f%%,  third=%.1f%%\n',  
pre_tariff_structure);  
fprintf('    after:   export=%.1f%%,  produce=%.1f%%,  third=%.1f%%\n',  
post_tariff_structure);  
fprintf('    percentage up%.1f\n', post_tariff_structure(2)-pre_tariff_structure(2));
```

```
pre_tariff_usshare = 100 - mean(jp_share_total(1:4));
```

```
post_tariff_usshare_pred = 100 - mean(future_predictions(:,6));
usshare_change = post_tariff_usshare_pred - pre_tariff_usshare;

pre_tariff_usvol = mean(us_brand_k(1:4));
post_tariff_usvol_pred = mean(future_predictions(:,5));
usvol_change = (post_tariff_usvol_pred - pre_tariff_usvol) / pre_tariff_usvol * 100;

fprintf('\n3. industry:\n');
fprintf('    - us percentage: +%.1f\n', usshare_change);
fprintf('    - us sales: +%.1f%%\n', usvol_change);

function [s1, s2, s3] = predict_shares(D, E, beta1, beta2)
    logit1 = beta1(1) + beta1(2)*D + beta1(3)*E;
    logit2 = beta2(1) + beta2(2)*D + beta2(3)*E;

    s3 = 1 / (1 + exp(logit1) + exp(logit2));
    s1 = exp(logit1) * s3;
    s2 = exp(logit2) * s3;
end
```

Question three code 1:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import statsmodels.formula.api as smf
from statsmodels.tsa.statespace.sarimax import SARIMAX

df = pd.read_csv("capacity_util.csv")
df = df.sort_values(['year','quarter']).reset_index(drop=True)

df['t'] = np.arange(1, len(df)+1)
df['quarter_label'] = df['year'].astype(str) + 'Q' + df['quarter'].astype(str)

sub_y, sub_q = 2022, 3
df['subsidy_dummy'] = df.apply(lambda r: int((r['year']>sub_y) or (r['year']==sub_y
and r['quarter']>=sub_q)), axis=1)
```

```
sub_start_idx = df.index[(df['year']==sub_y) & (df['quarter']==sub_q)][0]
df['post_sub_trend'] = df['t'] - sub_start_idx
df['post_sub_trend'] = df['post_sub_trend'].apply(lambda x: max(0,x))

tar_y, tar_q = 2025, 2
df['tariff_dummy'] = df.apply(lambda r: int((r['year']>tar_y) or (r['year']==tar_y and
r['quarter']>=tar_q)), axis=1)
tariff_start_idx = df.index[(df['year']==tar_y) & (df['quarter']==tar_q)][0]
df['post_tariff_trend'] = df['t'] - tariff_start_idx
df['post_tariff_trend'] = df['post_tariff_trend'].apply(lambda x: max(0,x))

for q in [2,3,4]:
    df[f'Q{q}'] = (df['quarter']==q).astype(int)

formula = "caputil ~ t + subsidy_dummy + post_sub_trend + tariff_dummy +
post_tariff_trend + Q2 + Q3 + Q4"
ols_res = smf.ols(formula, data=df).fit(cov_type='HAC', cov_kwds={'maxlags':4})
df['fitted_its'] = ols_res.fittedvalues

df_cf = df.copy()
post_tariff_mask = df_cf['t'] >= tariff_start_idx
df_cf.loc[post_tariff_mask, 'tariff_dummy'] = 0
df_cf.loc[post_tariff_mask, 'post_tariff_trend'] = 0
df_cf['fitted_no_tariff'] = ols_res.predict(df_cf)

last_year, last_quarter = df['year'].iloc[-1], df['quarter'].iloc[-1]
forecast_quarters = []
forecast_years = []
y, q = last_year, last_quarter
while y < 2026 or (y==2026 and q<=3):
    q += 1
    if q>4:
        q = 1
        y += 1
    forecast_years.append(y)
```



```
forecast_quarters.append(q)

n_forecast = len(forecast_quarters)

sar = SARIMAX(df['caputil'],
              order=(1,1,1),
              seasonal_order=(0,1,1,4),
              enforce_stationarity=False,
              enforce_invertibility=False)
sar_res = sar.fit(dispatch=False)
pred = sar_res.get_forecast(steps=n_forecast)
forecast_values = pred.predicted_mean.values

forecast_df = pd.DataFrame({
    'year': forecast_years,
    'quarter': forecast_quarters,
    'quarter_label': [f'{y}Q{q}' for y,q in zip(forecast_years, forecast_quarters)],
    'forecast_sarimax': forecast_values
})

def plot_series_with_actual(x, actual, y_model, ylabel, title, color='b', linestyle='--',
policy_lines=None):
    plt.figure(figsize=(14,5))
    plt.plot(x[:len(actual)], actual, color='k', linestyle='-', label='Actual')
    plt.plot(x, y_model, color=color, linestyle=linestyle, label=title)
    if policy_lines:
        for idx, label in policy_lines:
            plt.axvline(idx, color='gray', linestyle=':', label=label)
    plt.xticks(rotation=45)
    plt.xlabel('Quarter', fontsize=16, fontname='Times New Roman')
    plt.ylabel(ylabel, fontsize=16, fontname='Times New Roman')
    plt.title(title, fontsize=16, fontname='Times New Roman')
    plt.legend(fontsize=14)
    plt.tight_layout()
    plt.show()
```

```
plot_series_with_actual(df['quarter_label'], df['caputil'], df['fitted_its'],
                        'Capacity Utilization (%)', 'ITS Fitted vs Actual',
                        color='b', linestyle='--',
                        policy_lines=[(sub_start_idx,'CHIPS
Act'),(tariff_start_idx,'Tariff')])

plot_series_with_actual(df['quarter_label'], df['caputil'], df_cf['fitted_no_tariff'],
                        'Capacity Utilization (%)', 'Counterfactual (No Tariff) vs
Actual',
                        color='r', linestyle='--',
                        policy_lines=[(sub_start_idx,'CHIPS
Act'),(tariff_start_idx,'Tariff')])

full_x = list(df['quarter_label']) + list(forecast_df['quarter_label'])
full_y_model = list(df['caputil']) + list(forecast_df['forecast_sarimax'])
plot_series_with_actual(full_x, list(df['caputil']), full_y_model,
                        'Capacity Utilization (%)', 'SARIMAX Forecast vs
Actual',
                        color='g', linestyle='-.',
                        policy_lines=[(sub_start_idx,'CHIPS
Act'),(tariff_start_idx,'Tariff')])
```

Question three code 2:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import statsmodels.formula.api as smf
from statsmodels.tsa.statespace.sarimax import SARIMAX
import warnings

warnings.filterwarnings("ignore")
df = pd.read_csv('data.csv', dtype={'time': str})
df[['year', 'month']] = df['time'].str.split('.', expand=True).astype(int)
df['ym'] = df['year'].astype(str) + '-M' + df['month'].astype(str).str.zfill(2)
df = df.sort_values(['year', 'month']).reset_index(drop=True)
```

```
df['t'] = np.arange(1, len(df) + 1)
sub_y, sub_m = 2022, 8
tar_y, tar_m = 2025, 4
after_or_eq = lambda y, m, yy, mm: (y > yy) or (y == yy and m >= mm)
df['subsidy_dummy'] = df.apply(lambda r: 1 if after_or_eq(r['year'], r['month'], sub_y,
sub_m) else 0, axis=1)
df['tariff_dummy'] = df.apply(lambda r: 1 if after_or_eq(r['year'], r['month'], tar_y,
tar_m) else 0, axis=1)
sub_idx = df.index[(df['year'] == sub_y) & (df['month'] == sub_m)][0]
tar_idx = df.index[(df['year'] == tar_y) & (df['month'] == tar_m)][0]
df['post_sub_trend'] = (df['t'] - df.loc[sub_idx, 't']).apply(lambda x: max(0, x))
df['post_tar_trend'] = (df['t'] - df.loc[tar_idx, 't']).apply(lambda x: max(0, x))

series_info = [
    ('high_export', 'High-end exports'),
    ('mid_export', 'Mid-end exports'),
    ('low_export', 'Low-end exports')
]
forecast_dict = {}
for col, label in series_info:
    s = df.copy()
    s['y'] = s[col].astype(float)

    formula = "y ~ t + subsidy_dummy + post_sub_trend + tariff_dummy +
post_tar_trend"
    ols_res = smf.ols(formula, data=s).fit(cov_type='HAC', cov_kwds={'maxlags':
12})
    s['fitted_its'] = ols_res.fittedvalues
    s['subsidy_dummy_cf'] = 0
    s['tariff_dummy_cf'] = 0
    s['post_sub_trend_cf'] = 0
    s['post_tar_trend_cf'] = 0
    formula_cf = "y ~ t + subsidy_dummy_cf + post_sub_trend_cf + tariff_dummy_cf
+ post_tar_trend_cf"
    ols_cf_res = smf.ols(formula_cf, data=s).fit()
```

```
s['counterfactual'] = ols_cf_res.fittedvalues

last_y, last_m = s['year'].iloc[-1], s['month'].iloc[-1]
future = []
fy, fm = last_y, last_m
while (fy < 2026) or (fy == 2026 and fm < 7):
    fm += 1
    if fm > 12: fm = 1; fy += 1
    future.append((fy, fm))
steps = len(future)
sar = SARIMAX(s['y'], order=(1, 1, 1), seasonal_order=(0, 1, 1, 12),
              enforce_stationarity=False, enforce_invertibility=False)
sar_res = sar.fit(dispatch=False)
if steps > 0:
    pred = sar_res.get_forecast(steps=steps)
    pred_mean = pred.predicted_mean.values
    df_future = pd.DataFrame({'year': [y for y, m in future],
                             'month': [m for y, m in future],
                             'col': pred_mean})
    forecast_dict[col] = pd.concat([s[['year', 'month', col]], df_future],
    ignore_index=True)
else:
    forecast_dict[col] = s[['year', 'month', col]].copy()

plt.figure(figsize=(12, 5))
plt.plot(s['ym'], s['y'], '-', color='black', linewidth=2, label='Actual')
plt.plot(s['ym'], s['fitted_its'], '--', color='#DEEBF7', linewidth=2, label='ITS
Fitted')
fdf = forecast_dict[col]
fdf['ym'] = fdf['year'].astype(str) + '-M' + fdf['month'].astype(str).str.zfill(2)
plt.plot(fdf['ym'], fdf[col], '-', color='green', label='SARIMAX Forecast')
plt.axvline(f'{tar_y}-M{tar_m:02d}', color='purple', linestyle=':', linewidth=1)
plt.axvline(f'{sub_y}-M{sub_m:02d}', color='orange', linestyle=':', linewidth=1)
plt.xlabel('Month');
plt.ylabel('USD');
```

```
plt.title(label)
plt.xticks(rotation=45)
plt.legend()
plt.show()
```

Question four code:

```
import pandas as pd
import numpy as np
from statsmodels.tsa.holtwinters import ExponentialSmoothing
import matplotlib.pyplot as plt

df = pd.DataFrame({
    'observation_date': ['2020Q1', '2020Q2', '2020Q3', '2020Q4',
                        '2021Q1', '2021Q2', '2021Q3', '2021Q4',
                        '2022Q1', '2022Q2', '2022Q3', '2022Q4',
                        '2023Q1', '2023Q2', '2023Q3', '2023Q4',
                        '2024Q1', '2024Q2', '2024Q3', '2024Q4',
                        '2025Q1', '2025Q2'],
    'imports': [584.848, 524.345, 628.384, 667.804,
                655.896, 723.729, 755.133, 798.218,
                810.979, 878.702, 863.890, 819.332,
                770.471, 793.806, 803.765, 800.429,
                776.374, 834.137, 875.714, 870.587,
                970.776, 839.809],
    'tariff_revenue': [79.515, 55.361, 66.400, 73.230,
                      80.734, 89.810, 89.231, 96.631,
                      107.370, 110.217, 101.651, 90.093,
                      84.420, 83.508, 79.518, 79.084,
                      82.342, 78.941, 85.865, 87.199,
                      96.965, 267.681]
})

df['effective_tariff'] = df['tariff_revenue'] / df['imports']

imports_series = pd.Series(df['imports'].values, index=pd.period_range('2020Q1',
'2025Q2', freq='Q'))
```

```
model = ExponentialSmoothing(imports_series, seasonal='add',
seasonal_periods=4)
fit = model.fit()
forecast_future = fit.forecast(8)
t_future = df['effective_tariff'].iloc[-1]
t_future = df['effective_tariff'].iloc[-1]
epsilon_short = 0.02
epsilon_mid = 0.2
short_import = forecast_future.iloc[0:3] * (1 - epsilon_short)
short_revenue = t_future * short_import
mid_import = forecast_future.iloc[3:] * (1 - epsilon_mid)
mid_revenue = t_future * mid_import

imports_all = list(df['imports']) + list(short_import) + list(mid_import)
revenue_all = list(df['tariff_revenue']) + list(short_revenue) + list(mid_revenue)
x_labels_all = list(df['observation_date']) + list(forecast_future.index.astype(str))

def plot_dual_axis_with_forecast(x_labels, y1, y2, predict_start,
                                y1_label='Imports (Billion USD)',
                                y2_label='Tariff Revenue (Billion
USD)',
                                title='US Imports and Tariff Revenue'):

    fig, ax1 = plt.subplots(figsize=(12, 6))

    color1 = 'tab:blue'
    ax1.set_xlabel('Quarter')
    ax1.set_ylabel(y1_label, color=color1)
    ax1.plot(range(predict_start), y1[:predict_start], color=color1, marker='o',
linestyle='-', label=y1_label)
    ax1.plot(range(predict_start - 1, len(y1)), y1[predict_start - 1:], color=color1,
marker='o', linestyle='--')
    ax1.tick_params(axis='y', labelcolor=color1)
```

```
ax1.set_xticks(range(len(x_labels)))
ax1.set_xticklabels(x_labels, rotation=45)

ax2 = ax1.twinx()
color2 = 'tab:red'
ax2.set_ylabel(y2_label, color=color2)
ax2.plot(range(predict_start), y2[:predict_start], color=color2, marker='s',
linestyle='-', label=y2_label)
ax2.plot(range(predict_start - 1, len(y2)), y2[predict_start - 1:], color=color2,
marker='s', linestyle='--')
ax2.tick_params(axis='y', labelcolor=color2)

fig.tight_layout()
plt.title(title)
plt.show()
plot_dual_axis_with_forecast(
    x_labels=x_labels_all,
    y1=imports_all,
    y2=revenue_all,
    predict_start=len(df),
    title='Actual + Forecast US Imports and Tariff Revenue'
)
```

Question five code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from dateutil.relativedelta import relativedelta
from scipy import stats
import os

DATA_PATH = "data.csv"
VARS = ['ImportsCH', 'IP', 'SOX', 'MfgEmp']
EXOG_VARS = ['Tariff', 'Retaliation']
LAGS = 1
ALPHA = 0.10
```

```
FORECAST_END = "2027-04-01"
```

```
def read_data(path):
```

```
    df = pd.read_csv(path)
```

```
    if "Time" not in df.columns:
```

```
        for c in df.columns:
```

```
            if "time" in c.lower():
```

```
                df = df.rename(columns={c: "Time"})
```

```
                break
```

```
df["Time"] = df["Time"].astype(str).str.strip()
```

```
df = df[df["Time"].str.len() >= 6]
```

```
df["year"] = df["Time"].str[:4].astype(int)
```

```
df["month"] = df["Time"].str[4:6].astype(int)
```

```
df["date"] = pd.to_datetime(df[["year", "month"]].assign(day=1))
```

```
rename = {}
```

```
for c in df.columns:
```

```
    s = c.lower()
```

```
    if "import" in s:
```

```
        rename[c] = "ImportsCH"
```

```
    elif "ip" in s or "industrial" in s:
```

```
        rename[c] = "IP"
```

```
    elif "sox" in s:
```

```
        rename[c] = "SOX"
```

```
    elif "emp" in s:
```

```
        rename[c] = "MfgEmp"
```

```
    elif "tariff" in s or "trade" in s:
```

```
        rename[c] = "Tariff"
```

```
df = df.rename(columns=rename)
```

```
for v in ["ImportsCH", "MfgEmp"]:
```



```
        if v in df.columns:
            df[v] = pd.to_numeric(df[v].astype(str).str.replace(",",""),
errors="coerce")

    if "IP" in df.columns:
        df["IP"] = pd.to_numeric(df["IP"], errors="coerce")

    if "SOX" in df.columns:
        df["SOX"] = pd.to_numeric(df["SOX"], errors="coerce")

    if "Tariff" in df.columns:
        def clean_pct(x):
            if pd.isna(x): return np.nan
            s = str(x).replace("%","").replace("%","")
            try: return float(s)/100
            except: return np.nan
        df["Tariff"] = df["Tariff"].apply(clean_pct)

    df["Retaliation"] = (df["date"] >= pd.to_datetime("2025-04-01")).astype(int)

    df = df.set_index("date").sort_index()

    return df

def build_VARX(df, vars_y, vars_x, p):
    Y = np.log(df[vars_y]).diff().dropna()

    X = df[vars_x].loc[Y.index]

    T, k = Y.shape
    m = X.shape[1]

    rows = T - p
    Y_dep = Y.values[p:]

    Z_y = []
```

```
    for t in range(p, T):
        row = []
        for lag in range(1, p+1):
            row.extend(Y.values[t-lag])
        Z_y.append(row)
    Z_y = np.array(Z_y)

    Z = np.hstack([Z_y, X.values[p:], np.ones((rows,1))])

    return Y_dep, Z, Y

def estimate_VARX(Y_dep, Z, k, p, m):
    Theta = np.linalg.solve(Z.T @ Z, Z.T @ Y_dep)
    U = Y_dep - Z @ Theta
    Sigma = (U.T @ U) / (U.shape[0]-1)

    invZZ = np.linalg.inv(Z.T @ Z)
    se = np.zeros_like(Theta)

    for j in range(k):
        cov_beta = invZZ * Sigma[j,j]
        se[:,j] = np.sqrt(np.diag(cov_beta))

    return Theta, se, U

def forecast_growth(Theta, last_Yg, exog_future, steps, k, p, m):
    hist = list(last_Yg[-p:])
    out = np.zeros((steps,k))

    for t in range(steps):
        Z_lag = np.hstack([hist[-lag] for lag in range(1,p+1)])
        z = np.hstack([Z_lag, exog_future[t], 1.0])
        yhat = z @ Theta
        out[t] = yhat
        hist.append(yhat)
```

return out

```
def bootstrap_levels(df, vars_y, growth_fcst, resid, nsim=1000, alpha=0.10):
```

```
    k = len(vars_y)
```

```
    T_future = growth_fcst.shape[0]
```

```
    last_levels = df[vars_y].iloc[-1].values
```

```
    sims = np.zeros((nsim, T_future, k))
```

```
    rng = np.random.default_rng(123)
```

```
    for i in range(nsim):
```

```
        noise = resid[rng.integers(0, resid.shape[0], T_future)]
```

```
        g = growth_fcst + noise
```

```
        for j in range(k):
```

```
            sims[i,:,j] = last_levels[j] * np.exp(np.cumsum(g[:,j]))
```

```
    median = np.percentile(sims, 50, axis=0)
```

```
    lo = np.percentile(sims, 100*alpha/2, axis=0)
```

```
    hi = np.percentile(sims, 100*(1-alpha/2), axis=0)
```

```
    return median, lo, hi
```

```
if __name__ == "__main__":
```

```
    df = read_data(DATA_PATH)
```

```
    Y_dep, Z, Yg = build_VARX(df, VARS, EXOG_VARS, LAGS)
```

```
    k = len(VARS); p = LAGS; m = len(EXOG_VARS)
```

```
    Theta, se, U = estimate_VARX(Y_dep, Z, k, p, m)
```

```
    print("\n=== TARIF & RETALIATION COEFFICIENTS ===")
```

```
    idx_exog = k*p
```

```
    for j,v in enumerate(VARS):
```

```
        tar = Theta[idx_exog,j]
```

```
ret = Theta[idx_exog+1,j]
print(f'{v}: Tariff={tar:.6f}, Retaliation={ret:.6f}')

last_date = df.index.max()
end = pd.to_datetime(FORECAST_END)
steps = (end.year - last_date.year)*12 + (end.month - last_date.month)

last_tariff = df["Tariff"].iloc[-1]
future_exog = np.column_stack([
    np.repeat(last_tariff, steps),
    np.ones(steps)
])

Yg_vals = np.log(df[VARs]).diff().dropna().values
growth_fcst = forecast_growth(Theta, Yg_vals, future_exog, steps, k, p, m)
med, lo, hi = bootstrap_levels(df, VARs, growth_fcst, U, nsim=1000,
alpha=ALPHA)

dates = [last_date + relativedelta(months=+i) for i in range(1,steps+1)]
f = pd.DataFrame(index=dates)
for i,v in enumerate(VARs):
    f[v+"_median"] = med[:,i]
    f[v+"_lo"] = lo[:,i]
    f[v+"_hi"] = hi[:,i]

f.to_csv("forecast_VARX.csv", float_format="%.4f")
print("\nForecast saved to forecast_VARX.csv")

for i,v in enumerate(VARs):
    plt.figure(figsize=(9,4))
    plt.plot(df.index, df[v], label="Historical")
    plt.plot(f.index, f[v+"_median"], "--", label="Forecast")
    plt.fill_between(f.index, f[v+"_lo"], f[v+"_hi"], alpha=0.25, label="90%
CI")

    plt.title(v)
```

```
plt.legend()  
plt.tight_layout()  
plt.show()
```

```
print("Done.")
```